BRAINSPARK

DFM Rule Book

Every rule the engine judges a part against, by commodity

247 RULES · 35 PROCESS FAMILIES · 6 COMMODITY GROUPS

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This book is generated from the catalogue the engine actually runs. Nothing on these pages was typed by hand, which means it cannot disagree with the tool: every threshold, comparison, citation and per-material override below is the value the analysis will use on your part.

Two figures travel with each rule that a checklist would leave out. The first is what the threshold rests on - a named standard, an industry consensus, or this tool's own cost model. The second is whether anyone has since gone and checked it. Where the answer is "not yet", the page says so.

123	103	21	152	29
HIGH SEVERITY	MEDIUM	LOW	MATERIAL-SPECIFIC THRESHOLDS	AUDITED TO A PRIMARY SOURCE

WHAT THIS BOOK IS NOT

It is not a design standard. Most of these thresholds are widely-published practice rather than a clause anyone here has read in the original document, and 2 are actively contested - those are named in Appendix B. Where your plant has its own number, it outranks everything in this book, and the tool lets you set it. It is also not a complete list of what matters: Appendix A sets out 18 things the engine deliberately does not claim to check.

THE SIX COMMODITY GROUPS

Castings
105 RULES 10 PROCESSES

High-pressure die casting (Al / Mg) - High-pressure die casting (Zinc) - Low-pressure die casting - Gravity die casting - Squeeze casting - Semi-solid casting (thixo / rheo) - Sand casting - Shell mould casting - Investment casting - Centrifugal casting

Sheet metal and forming
42 RULES 7 PROCESSES

Sheet metal / stamping - Deep drawing (multi-stage) - Fine blanking - Hot stamping / press hardening - Roll forming - Hydroforming - Metal spinning

Forging and bulk forming
27 RULES 5 PROCESSES

Forging (hot) - Forging (cold) - Open-die forging - Cold heading / upsetting - Extrusion

Machining and material removal
24 RULES 5 PROCESSES

Machining (CNC mill/turn) - Turning (CNC lathe) - Deep-hole / gun drilling - Broaching - Wire EDM

Polymers, rubber and composites
32 RULES 5 PROCESSES

Injection moulding - Rubber moulding - Thermoforming (vacuum / pressure) - Rotational moulding - Composite layup / RTM

Powder and additive
17 RULES 3 PROCESSES

Powder metallurgy (press & sinter) - Metal injection moulding (MIM) - Laser powder bed fusion (DMLS / SLM)

CHAPTER 0

How to read this book

A rule has three outcomes, not two

PASS	the measurement was taken and it is inside the limit.
FAIL	the measurement was taken and it is outside the limit. This is a finding.
NOT EVALUATED	the geometry needed for the measurement was not present or could not be read. The rule is reported with the reason, and it is NOT counted as a pass. A report that lists only failures invites you to assume everything else was checked.

Anatomy of an entry

```
wallP50Mm  in  1 to 3 mm
im-wall-thickness-range
```

The grey line is the test itself, in the engine's own terms: the name of the measurement taken from your CAD, the comparison, the threshold and its unit. Below it is the rule id, which is what appears in the analysis output and in the Excel export - quote it if you want to challenge a result. Everything after that is prose: why the limit exists, what to do about it, and where the number came from.

Which threshold applies to your part

The number printed at the head of an entry is the process-generic one. Many rules carry a stricter or looser value for a specific alloy or material family, and those are listed underneath. The engine resolves them in this order, and stops at the first match:

1. **Your company standard** a value set in your workspace. It outranks every published guideline, and the report credits it to you rather than to a handbook.
2. **The exact material grade** e.g. the row for DP980 rather than for steel generally.
3. **The material family** e.g. aluminium, titanium, thermoplastic.
4. **The process-generic value** the headline number in this book.

How "best first" was decided

Rules are ordered within every commodity by the composite below, and the leaderboard in Chapter 1 uses the same figure across the whole catalogue. It is printed here in full because a ranking with a hidden formula is an opinion in a lab coat - every term is something you can check from the entry itself.

```
3 x severity           high 3   medium 2   low 1
2 x source strength    named standard 3   industry consensus 2   own cost model 1
+ 2                    carries per-GRADE thresholds (a number tuned to your alloy)
+ 1                    carries per-FAMILY thresholds
+ 2                    the cost engine can put a currency figure on the finding
+ 2                    blocking (the part cannot be made by this route at all)
- 3                    the threshold is contested in the audit register
```

A high score means the rule is severe, well-sourced, tuned to your material and priced. It does not mean it will fire on your part - that depends on whether the geometry carries the feature it measures.

CONTENTS

By commodity

CHAPTER 1

The 40 highest-ranked rules

Across all 35 process families, by the composite on the previous page. If you read nothing else, read these: they are the severe, well-sourced, material-aware checks. Each is repeated in full in its own commodity chapter.

#	RULE	COMMODITY / PROCESS	TEST	SCORE
01	Wall thickness outside the practical moulding range	Injection moulding	1-3 mm	17
02	Rib too thick at its base for the wall it stands on	Injection moulding	<= 0.6 rib t / wall t	16
03	Too large to cold head from wire	Cold heading / upsetting	<= 150 mm largest dimension	15
04	More metal than a header can cut as a slug	Cold heading / upsetting	<= 80 cm3	15
05	Too deep to draw in one operation	Deep drawing (multi-stage)	<= 0.75 depth/width	15
06	Cup too deep for the drawing-ratio table	Deep drawing (multi-stage)	<= 0 beyond the table (1...	15
07	Too little metal around a fastener boss	High-pressure die casting (Al / M...	>= 2 boss dia / hole dia	15
08	Wall area below the draft NADCA computes for t...	High-pressure die casting (Al / M...	<= 5 % of wall area	15
09	Too little metal around a fastener boss	High-pressure die casting (Zinc)	>= 2 boss dia / hole dia	15
10	Wall area below the draft NADCA computes for z...	High-pressure die casting (Zinc)	<= 5 % of wall area	15
11	Wall area below the minimum draft angle	Injection moulding	<= 5 % of wall area	15
12	Tolerance tighter than injection moulding holds	Injection moulding	>= 0.2 mm total band	15
13	Too large for metal injection moulding	Metal injection moulding (MIM)	<= 100 mm largest dimension	15
14	Too much metal to debind and sinter	Metal injection moulding (MIM)	<= 30 cm3	15
15	Tolerance tighter than roll forming holds	Roll forming	>= 0.6 mm total band	15
16	Strip utilisation below the industry target	Sheet metal / stamping	>= 70 % of strip area	15
17	Tolerance tighter than sheet metal / stamping holds	Sheet metal / stamping	>= 0.4 mm total band	15
18	Wall too thin for a slow-filled squeeze casting	Squeeze casting	>= 3 mm	15
19	Wall thinner than the die can hold	Extrusion	>= 1 mm	14
20	Section is not uniform enough for the die to run b...	Extrusion	<= 0.35 (p95-p5)/p50	14
21	Web too thin for the forming load available	Forging (cold)	>= 1.5 mm	14
22	Wall area below the minimum hot-forging draft	Forging (hot)	<= 5 % of wall area	14
23	Web too thin to forge	Forging (hot)	>= 3 mm	14
24	Tolerance tighter than forging holds	Forging (hot)	>= 1 mm total band	14
25	Wall area below the minimum gravity die-casting...	Gravity die casting	<= 5 % of wall area	14
26	Local section below the process minimum	Gravity die casting	>= 2.5 mm	14
27	Wall thickness outside the gravity die-casting range	Gravity die casting	3-8 mm	14
28	Non-uniform wall thickness	Gravity die casting	<= 0.7 (p95-p5)/p50	14
29	Cored hole beyond the core-pin slenderness limit	High-pressure die casting (Al / M...	<= 10 core L/D	14
30	Corner radius too tight for the forming pressure a...	Hydroforming	>= 3 r/t	14
31	Inside corner radius below the DuPont stress-con...	Injection moulding	>= 1 x required fillet	14
32	Local section below the process minimum	Investment casting	>= 1.2 mm	14
33	Tolerance tighter than investment casting holds	Investment casting	>= 1 x CT5-7 capability	14
34	Wall thickness outside the investment-casting ra...	Investment casting	1.5-6 mm	14
35	Thin web will chatter or deflect under the cutter	Machining (CNC mill/turn)	>= 1.5 mm	14
36	Tolerance tighter than machining holds	Machining (CNC mill/turn)	>= 0.05 mm total band	14
37	Wall area below the minimum sand-casting draft	Sand casting	<= 5 % of wall area	14
38	Local section below the process minimum	Sand casting	>= 3.5 mm	14
39	Tolerance tighter than sand casting holds	Sand casting	>= 1 x SFSA 2000 capability	14
40	Wall thickness outside the sand-casting range	Sand casting	4-12 mm	14

CHAPTER 2

Castings

Metal poured or injected into a mould. The shared enemies are draft, wall uniformity, section transitions and anything that traps the pattern or the core.

105 RULES · 10 PROCESSES · 38 HIGH / 51 MEDIUM / 16 LOW · 18 CARRY MATERIAL-SPECIFIC THRESHOLDS

High-pressure die casting (Al / Mg)

17 RULES

01 Too little metal around a fastener boss

HIGH

```
nadcaBossDiaToHoleDia  >=  2 boss dia / hole dia
hpdc-boss-wall
```

The shear strength and stiffness of a die-casting alloy are far below steel, so the wedging action of a thread dilates the boss — the hole grows, the mating threads lose contact and the joint fails. Metal around the hole is what resists it.

FIX Take the boss to at least twice the fastener diameter, or fit a steel insert and let the insert carry the thread.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §3.2 "Threaded Fasteners" p.51, read first-hand: "the boss diameter should be at least twice the bolt diameter".

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

02 Wall area below the draft NADCA computes for this alloy and depth

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
hpdc-draft-minimum  ·  cutoff 1 deg
```

Aluminium shrinks onto the die steel as it solidifies, so a die casting needs more draft than a moulding — and the amount is not a constant. It falls as the feature gets deeper and it changes with the alloy, so one figure for a whole catalogue is right at one depth only. Insufficient draft galls the die surface and shortens die life as well as risking ejector distortion.

FIX Take the drawing to the general-note draft the analysis prints for this part, and call the shallow features out separately — a shallow feature needs MORE draft, not less.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §3.1 "Draft requirements" pp.45-46, READ FIRST-HAND. A = 57.2738 / (C·sqrt(L)), L in inches. Draft constants C by alloy group and surface: zinc and ZA 50 inside / 100 outside / 34 hole total; magnesium 35/70/24; aluminium 30/60/20; copper 25/50/17. Verified against the book's own worked example for an aluminium inside surface (0.1 in -> 6.0 deg, 1.0 in -> 1.9 deg, 5.0 in -> 0.85 deg). Judged here at the OUTSIDE constant, the lower of the two, because the engine cannot yet separate an ejector-half surface from a cover-half one — so this UNDER-reports on inside walls, which need twice as much.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

03 Cored hole beyond the core-pin slenderness limit

HIGH

```
maxHoleDepthToDia  <=  10 core L/D
hpdc-core-ld
```

A cored hole is a steel pin standing in the die with molten aluminium injected around it at speed. Beyond about 10:1 the pin deflects under that pressure, the hole drifts, and the pin becomes a consumable that breaks mid-run and stops the cell.

FIX Shorten the cored depth and drill the rest, open the diameter, or core from both ends so each pin is half the length.

SOURCE Die-casting design guidance (core pin L/D <= 10:1 for aluminium).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · CORROBORATED BY SEARCH

BY MATERIAL FAMILY

<= 12 magnesium

04 Undercuts require slides or lifters in the die

HIGH

```
undercutFaceCount  <=  0 regions
hpdc-undercuts
```

A die-cast undercut needs a slide or a loose core. Both add die cost and maintenance, and a loose core adds a manual handling step to every shot.

FIX Reorient the part to the draw, move the parting line, or price the slide explicitly.

SOURCE Die casting tool design: the die opens along one axis, so a re-entrant feature needs a slide or a lifter — a cam mechanism, its wear parts and the cycle time to move it. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Wall thickness outside the die-casting range

HIGH

```
wallP50Mm  in  1 to 3.5 mm
hpdc-wall-thickness-range  ·  PRICED BY THE COST ENGINE
```

Below about 1 mm the die will not fill reliably and cold shuts appear. Above about 3.5 mm the extra metal is not load-bearing metal: a die casting grows a dense chilled skin only 0.4 to 0.5 mm deep, and that skin does NOT get thicker as the wall does — so a heavy section is the same strong skin plus a porous core, and it holds the cycle open while it freezes.

FIX Hold a uniform nominal wall in the 2.0–3.5 mm band and core out heavy sections.

SOURCE Aluminium HPDC design guidance: 1.0 mm practical minimum for a small part, 2.0-3.5 mm the usual band. INDUSTRY CONSENSUS and deliberately NOT promoted: NADCA Product Design for Die Casting 7th ed. §3.1, read first-hand, states there are no hard and fast rules governing maximum and minimum wall thickness and gives no table. The physical argument for the upper bound IS from that book, §4.2 pp.93-94 citing Borland and Tsumagari (2006): the chilled skin is 0.38-0.50 mm and does not thicken with the wall, so the extra metal in a heavy section is porous core. Table 4.8 p.100 rates each alloy for die-filling capacity, which is the one axis on which the minimum genuinely varies.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · CONTESTED - SEE APPENDIX B

BY MATERIAL FAMILY

1.5-4	aluminium	**1.3-3.5**	magnesium

06 Cored hole below the draft NADCA computes for its depth

MEDIUM

```
coredHoleDraftPerSideDeg  >=  2 deg per side
hpdc-cored-hole-draft
```

A casting shrinks ONTO a core pin rather than away from it, so a cored hole needs more draft than an outside wall. How much more depends on how deep the hole is: NADCA asks about 4.2 degrees per side on a 3 mm bore in aluminium and about 1.0 on a 50 mm one. A pin that binds galls, deflects and puts the hole off position; a bound pin in a deep hole can pull the casting off the ejector side.

FIX Draft the bore to the per-side figure the analysis prints for its measured depth. Where the hole must be parallel, drill it after casting and cost the secondary operation rather than fighting the pin.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §3.1 pp.45-46, READ FIRST-HAND. The "Hole, Total" column gives C = 20 for aluminium, 24 magnesium, 34 zinc and ZA, 17 copper, and is the TOTAL included angle — so the per-side requirement is half of what the formula returns. Each bore's depth comes from its own measured wall area and radius, and the worst bore on the part sets the margin.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

07 Inside corner radius smaller than the wall it joins

MEDIUM

```
nadcaFilletToWall  >=  1 R / wall t
hpdc-internal-radius
```

A sharp inside corner concentrates stress in the part AND in the die that forms it, and the die is the one that fails first — the corner is where heat checking starts. It also restricts metal flow into the corner, so the region fills last and fills badly.

FIX Open the inside radius to at least one wall thickness, and carry the matching outside radius at R + t so the wall stays uniform round the corner.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), Figure 3.2 pp.40-41, read first-hand: at a junction of uniform walls the optimal inside radius is R1 >= T1 with R2 = R1 + T1; between non-uniform walls R1 >= 2/3(T1 + T2). R1 = 0 is listed as "not recommended - will cause weak part & die casting die".

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

08 Machining stock deep enough to cut through the chilled skin

MEDIUM

```
nadcaMachiningStockToSkinMm  <=  0.5 mm removed
hpdc-machining-stock-skin
```

A die casting solidifies against cold steel, so it grows a dense, fine-grained skin about 0.4 to 0.5 mm thick that carries most of its strength and all of its pressure tightness. The skin does NOT get thicker as the wall does. Cutting deeper than it exposes the porous core — which is why a machined sealing face leaks when an as-cast one would not.

FIX Hold machining stock at or below 0.5 mm on faces that must seal or carry load, or specify a vacuum or squeeze process where the core is dense enough to cut into.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §4.2 pp.93-94 and Figure 4.6, read first-hand, citing Borland and Tsumagari (2006): the skin is 0.38-0.50 mm, is not a function of wall thickness, and "removal of the skin to a depth greater than 0.020 in. (0.50 mm) by secondary processes such as machining, increases the chance of exposing porosity in the core". NOTE A CONTRADICTION INSIDE THE BOOK: §3.4 and the glossary both give the same skin as 0.020 in. but convert it to "0.8 mm", which is arithmetically wrong — 0.020 in. is 0.508 mm. The IMPERIAL figure is 0.020 in. in all three places, so 0.5 mm is the correct metric value and the two 0.8 mm figures are the book mis-converting. Recorded here so the threshold is not "corrected" to 0.8 by a later reader.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

09 Hole too small to be cast — it will be drilled

MEDIUM

```
minHoleDiaMm  >=  2.5 mm dia
hpdc-min-cored-hole
```

The hole is made by a steel pin standing in the die, taking the full shot pressure on every cycle. Below roughly 2.5 mm in aluminium the pin is too slender to survive production — it soldiers, bends and eventually snaps — and unsupported pins under about 1.5 mm are not run at all. The feature does not disappear from the part; it moves to the drill, and a drilling operation nobody costed is a piece price nobody predicted.

FIX Open the hole to at least 2.5 mm to cast it, or delete it from the casting and quote it as a drilled feature so the secondary operation is in the price from the start.

SOURCE Die casting core-pin design guidance: unsupported core pins below about 1.5 mm are avoided or reinforced; the practical as-cast aluminium hole floor sits above that at roughly 2.5 mm. Consistent across supplier design guides; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 2 Magnesium AZ91D (die-cast)

10 Rib heavier than the wall it stands on

MEDIUM

```
maxRibThicknessToWall  <=  0.8 rib t / wall t
```

hpdc-rib-thickness-max · PRICED BY THE COST ENGINE

A rib approaching the full wall thickness makes the junction the thickest section in the casting. That is where shrinkage porosity collects, and it holds the die open while it solidifies.

FIX Bring the rib base to roughly 60–80% of the adjoining wall and fillet the root rather than thickening it.

SOURCE Die-casting design guidance (rib base ~60–80% of the adjoining wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

11 Tolerance tighter than NADCA Standard capability at that dimension

MEDIUM

```
nadca402ToleranceMargin  >=  1 x S-4A-1 capability
```

hpdc-tolerance-capability

As-cast capability grows with the dimension: NADCA promises a total band of 0.5 mm on a 25 mm feature and adds 0.05 mm of band per additional inch. A band tighter than the table means machining after casting - a fixture, a datum scheme and a second cost centre - or Precision-grade tooling, which is a die-cost decision, not a default.

FIX Open the band to the S-4A-1 value for that dimension, specify Precision grade where the fit genuinely needs it, or machine the feature and cost the operation.

SOURCE NADCA #402, Product Specification Standards for Die Castings, 2021 (11th ed.), Tables S-4A-1-21 and P-4A-1-21 pp.4A-7/8, READ FIRST-HAND: aluminium, magnesium and zinc ±0.25 mm for the first 25.4 mm plus ±0.025 mm per additional 25.4 mm (Standard); ±0.05 mm base (Precision); copper ±0.36/±0.076. Verified against the standard's own worked example - a 127 mm aluminium dimension carries ±0.35 mm Standard, ±0.15 mm Precision.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

12 Rib taller than the wall will fill and eject

MEDIUM

```
maxRibHeightToWall  <=  3 rib h / wall t
```

hpdc-rib-height

A deep rib is a deep, narrow blade of die steel. It runs hotter than the rest of the die, heat-checks first, and is the feature most likely to break out in service; the casting also grips it hard on ejection.

FIX Limit rib height to about 3x the wall, or split one deep rib into several shallower ones and increase the draft on what remains.

SOURCE Die-casting design guidance (rib height <= 3x wall; deep ribs need 3–5 deg draft).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 4 magnesium

13 Rib too thin to fill before it freezes

MEDIUM

```
minRibThicknessToWall  >=  0.6 rib t / wall t
```

hpdc-rib-thickness-min

Aluminium loses superheat fast in a thin section. A rib much below 60% of the wall chills before the cavity is full and leaves a cold shut at the rib tip — a crack starter, not just a cosmetic defect.

FIX Thicken the rib toward 60–80% of the wall, or shorten it so the flow path to its tip is shorter.

SOURCE Die-casting design guidance (rib base ~60–80% of the adjoining wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 0.5 magnesium

14 Non-uniform wall thickness

MEDIUM

```
wallSpreadRatio  <=  0.6 (p95-p5)/p50
hpdc-wall-uniformity
```

A heavy section next to a thin one is the last place to freeze and has no feed path once the gate has solidified, so it draws shrinkage porosity. It also distorts the casting as the two sections contract at different times.

FIX Core out the heavy sections to a uniform nominal wall and blend the transitions rather than stepping them.

SOURCE Die-casting design guidance (uniform section thickness; heavy sections trap shrinkage porosity).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

15 Surface finish tighter than the process holds over die life

LOW

```
nadcaRoughnessMargin  >=  1 x process capability
hpdc-as-cast-finish
```

A die holds its best finish when it is new and loses it as it heat-checks. Specifying the new-die figure is not a specification, it is a die-replacement schedule — and the die caster will price it as one.

FIX Specify the over-die-life figure, or accept a periodic die refurbishment and price it into the tooling amortisation.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), Table 3.26 p.68, read first-hand: aluminium, ZA-12 and ZA-27 hold 63 microinch or better in a new die and 100-125 over its life; magnesium 63 and 63; zinc and ZA-8 32 and 63.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

16 Flatness tighter than the as-cast table holds

LOW

```
nadca402FlatnessMargin  >=  1 x S-4A-8 capability
hpdc-flatness-capability
```

As-cast flatness is 0.20 mm over the first 76 mm of surface and grows 0.08 mm per additional inch (Standard). A flatness callout tighter than that is a straightening or machining operation, and the standard's own guidance is that sinks under large bosses and uneven wall heights are what break it.

FIX Open the flatness to the S-4A-8 value for the surface's largest dimension, specify Precision grade, or machine the face and respect the 0.5 mm skin.

SOURCE NADCA #402, Product Specification Standards for Die Castings, 2021 (11th ed.), Tables S-4A-8-21 and P-4A-8-21 pp.4A-23/24, READ FIRST-HAND: Standard 0.20 mm to 76.2 mm largest dimension + 0.08 mm per additional 25.4 mm; Precision 0.13 + 0.05. Verified against the standard's own worked example - a 254 mm diagonal carries 0.76 mm Standard, 0.48 mm Precision.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

17 Inside corner below the smallest radius a die is cut to

LOW

```
minInternalCornerRadiusMm  >=  0.8 mm
hpdc-internal-radius-floor
```

Below about 0.8 mm the corner stops being a radius the toolmaker cuts and becomes an edge the cutter leaves, which heat-checks early however thin the wall is.

FIX Carry a physical radius on every inside corner, even where the wall is thin enough that the ratio rule is satisfied.

SOURCE Die-casting tool practice: an inside corner is formed by an external corner on the die, and a cutter leaves a radius. Industry consensus, no primary source audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

High-pressure die casting (Zinc)

18 RULES

01 Too little metal around a fastener boss

HIGH

```
nadcaBossDiaToHoleDia  >=  2 boss dia / hole dia
hpdc-zinc-boss-wall
```

The wedging action of a thread dilates a die-cast boss — the hole grows, the mating threads lose contact and the joint fails. Zinc is the softest of the die-casting alloys, so it dilates the most readily of any of them.

FIX Take the boss to at least twice the fastener diameter, or fit an insert and let the insert carry the thread.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §3.2 "Threaded Fasteners" p.51, read first-hand: "the boss diameter should be at least twice the bolt diameter".

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

02 Wall area below the draft NADCA computes for zinc at this depth

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
hpdc-zinc-draft-minimum  ·  cutoff 0.5 deg
```

Zinc shrinks less onto the die than aluminium does and is cast at a lower temperature, so it needs the least draft of the four groups NADCA tabulates — its outside constant is 100 against aluminium's 60. That is a ratio of 0.6, not a licence for zero draft: the requirement still climbs as the feature gets shallower.

FIX Take the drawing to the general-note draft printed for this part. Zinc tolerates the least draft of any die-casting alloy, so where a feature cannot carry it, zinc is the alloy most likely to make it work.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §3.1 "Draft requirements" pp.45-46, READ FIRST-HAND. A = 57.2738 / (C·sqrt(L)), L in inches. Draft constants C by alloy group and surface: zinc and ZA 50 inside / 100 outside / 34 hole total; magnesium 35/70/24; aluminium 30/60/20; copper 25/50/17. Verified against the book's own worked example for an aluminium inside surface (0.1 in -> 6.0 deg, 1.0 in -> 1.9 deg, 5.0 in -> 0.85 deg). Judged here at the OUTSIDE constant, the lower of the two, because the engine cannot yet separate an ejector-half surface from a cover-half one — so this UNDER-reports on inside walls, which need twice as much.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

03 Local section below the process minimum

HIGH

```
wallP5Mm  >=  0.5 mm
hpdc-zinc-min-section
```

Zinc fills thinner than any other die-cast alloy, and 0.5 mm is roughly where even zinc stops. A local web thinner than the nominal wall is where a short shot starts.

FIX Thicken the thinnest region, or move the feature to where the section is already full.

SOURCE Process minimum-section guidance. Checked on the 5th-percentile wall, which is the thinnest measured place rather than the nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Wall thickness outside the zinc die-casting range

HIGH

```
wallP50Mm  in  0.6 to 3 mm
hpdc-zinc-wall-thickness-range
```

Zinc runs at a far lower melting point than aluminium in a hot-chamber machine, so it fills sections aluminium cannot — 0.6 mm is routine and 0.4 mm is achievable on small parts. That thinness is the reason to choose zinc at all. Above about 3 mm the section is heavy, slow to freeze and porous, and the material cost of a dense alloy starts to dominate.

FIX Take advantage of the alloy: thin the nominal wall towards 1.0-1.5 mm and add ribs for stiffness. A zinc part left at an aluminium wall is paying for metal it does not need.

SOURCE Zinc hot-chamber die-casting design guidance (ZAMAK): 0.6 mm practical minimum, 1.0-2.0 mm typical nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Cored hole below the draft NADCA computes for zinc at its depth

MEDIUM

```
coredHoleDraftPerSideDeg  >=  1 deg per side
hpdc-zinc-cored-hole-draft
```

Zinc grips a core pin less than aluminium does — NADCA gives it a hole constant of 34 against aluminium's 20 — but the pin still has to release, and the requirement still rises as the bore gets shallower.

FIX Draft the bore to the per-side figure printed for its measured depth, or drill after casting.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §3.1 pp.45-46, READ FIRST-HAND. The "Hole, Total" column gives C = 20 for aluminium, 24 magnesium, 34 zinc and ZA, 17 copper, and is the TOTAL included angle — so the per-side requirement is half of what the formula returns. Each bore's depth comes from its own measured wall area and radius, and the worst bore on the part sets the margin.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

06 Inside corner radius smaller than the wall it joins

MEDIUM

```
nadcaFilletToWall  >=  1 R / wall t
hpdc-zinc-internal-radius
```

A sharp inside corner is a stress raiser in the part and a heat-check origin in the die. Zinc runs cooler than aluminium so the die lasts longer, but the corner is still where it starts to fail.

FIX Open the inside radius to at least one wall thickness, and carry the outside radius at R + t.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), Figure 3.2 pp.40-41, read first-hand: R1 >= T1 at a uniform wall junction, R2 = R1 + T1; R1 = 0 is "not recommended - will cause weak part & die casting die".

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

07 Machining stock deep enough to cut through the chilled skin

MEDIUM

```
nadcaMachiningStockToSkinMm  <=  0.5 mm removed
hpdc-zinc-machining-stock-skin
```

The dense chilled skin is about 0.4 to 0.5 mm thick and does not get thicker as the wall does. Cutting deeper exposes the porous core, which is why a machined sealing face leaks where an as-cast one would not.

FIX Hold machining stock at or below 0.5 mm on faces that must seal or carry load.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §4.2 pp.93-94 and Figure 4.6, read first-hand, citing Borland and Tsumagari (2006).

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

08 Tolerance tighter than NADCA Standard capability at that dimension

MEDIUM

```
nadca402ToleranceMargin  >=  1 x S-4A-1 capability
hpdc-zinc-tolerance-capability
```

Zinc shares the aluminium row of the NADCA linear table - a total band of 0.5 mm on the first 25.4 mm, Standard grade. Zinc's real advantage is on the Precision side: the standard notes tighter figures are attainable with artificial ageing, and miniature zinc machines reach tighter still (Section 4B).

FIX Open the band to the S-4A-1 value for that dimension, or specify Precision grade - which for zinc may need artificial ageing to hold through the alloy's creep.

SOURCE NADCA #402, Product Specification Standards for Die Castings, 2021 (11th ed.), Tables S-4A-1-21 and P-4A-1-21 pp.4A-7/8, READ FIRST-HAND. The zinc column equals the aluminium one at Standard grade; the P-4A-1 notes record that artificial ageing and miniature die casting reach tighter than the table.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

09 Boss taller than the core pin can hold

MEDIUM

```
maxBossHeightToDia  <=  4 height/dia
hpdc-zinc-boss-height
```

A tall boss is cored by a pin standing in the flow, and the taller it is the more it deflects and the sooner it erodes. Zinc is kinder to pins than aluminium but the geometry still governs.

FIX Shorten the boss, widen it, or support it with ribs to its base.

SOURCE Die casting boss guidance: a boss is cored by a pin standing in the die and the same slenderness argument applies as to a cored hole, with zinc holding the finest pins of the die-casting alloys. The 4:1 figure sits above the aluminium core-pin limit used elsewhere in this catalogue because a boss pin is supported by the surrounding cavity where a through-core pin is not. DERIVED from that ordering; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

10 Cored hole beyond the core-pin slenderness limit

MEDIUM

```
maxHoleDepthToDia  <=  12 core L/D
hpdc-zinc-core-ld
```

A cast hole is made by a steel pin standing in the flow. Zinc enters cooler than aluminium, so the pin survives a longer reach before it bends or erodes, but a slender pin is still the first thing in the die to fail.

FIX Shorten the cored hole, open its diameter, or cast it partly and drill the rest.

SOURCE Zinc die-casting design guidance (core-pin length-to-diameter limits).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

11 Hole too small to be cast — it will be drilled

MEDIUM

```
minHoleDiaMm  >=  1.5 mm dia
hpdc-zinc-min-cored-hole
```

Zinc runs cooler and thinner than aluminium and is the least aggressive of the die-casting alloys on tooling, so a hot-chamber die will hold a finer core pin than an aluminium die will. It will not hold an arbitrarily fine one: below about 1.5 mm the pin is unsupported steel in a pressurised cavity and breaks.

FIX Open the hole to at least 1.5 mm, or move it to a drilling operation and cost that operation.

SOURCE Die casting core-pin design guidance: avoid unsupported cores below about 1.5 mm diameter, or reinforce them. Zinc is the alloy this floor is most nearly achievable in. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

12 Rib heavier than the wall it stands on

MEDIUM

```
maxRibThicknessToWall  <=  1 rib/wall
hpdc-zinc-rib-thickness-max
```

Zinc fills thin ribs readily, which is why zinc parts are ribbed rather than thick. A rib heavier than the wall wastes that advantage and slows the cycle.

FIX Thin the rib, or add two thinner ribs instead of one heavy one.

SOURCE Casting rib-proportion guidance (rib thickness against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

13 Rib too thin to fill before it freezes

MEDIUM

```
minRibThicknessToWall  >=  0.5 rib/wall
hpdc-zinc-rib-thickness-min
```

A rib is the last thing to fill and the first thing to freeze, because it presents the most surface for its volume. Below this fraction of the nominal wall it will not fill reliably.

FIX Thicken the rib towards the process minimum, or shorten it so the metal has less distance to travel.

SOURCE Casting rib-proportion guidance (fill limit against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

14 Undercuts require slides or lifters in the die

MEDIUM

```
undercutFaceCount  <=  0 regions
hpdc-zinc-undercuts
```

An undercut needs a slide or a loose core, which adds die cost and maintenance. Zinc dies carry slides more readily than aluminium dies because the alloy is far kinder to the tooling, so this is a cost question rather than a feasibility one.

FIX Re-orient the part on the parting line, or split the feature so it forms between the two die halves.

SOURCE Die-casting design guidance (side actions and loose cores).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

15 Non-uniform wall thickness

MEDIUM

```
wallSpreadRatio  <=  0.6 (p95-p5)/p50
hpdc-zinc-wall-uniformity
```

A heavy section beside a thin one is the last place to freeze and has no feed path once the gate solidifies, so it draws shrinkage porosity and distorts as the two sections contract at different times.

FIX Core out the heavy sections to a uniform nominal wall and blend the transitions rather than stepping them.

SOURCE Die-casting design guidance (uniform section thickness).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

16 Surface finish tighter than the process holds over die life

LOW

```
nadcaRoughnessMargin  >=  1 x process capability
hpdc-zinc-as-cast-finish
```

Zinc holds the best as-cast finish of the die-casting alloys — 32 microinch in a new die — but that degrades to about 63 over the life of the die. Specifying the new-die figure is a die-replacement schedule rather than a specification.

FIX Specify the over-die-life figure, or accept periodic die refurbishment and price it into the tooling.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), Table 3.26 p.68, read first-hand: zinc and ZA-8 hold 32 microinch or better in a new die and 63 over its life.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

17 Flatness tighter than the as-cast table holds

LOW

```
nadca402FlatnessMargin  >=  1 x S-4A-8 capability
hpdc-zinc-flatness-capability
```

As-cast flatness is 0.20 mm over the first 76 mm of surface and grows 0.08 mm per additional inch (Standard). A flatness callout tighter than that is a straightening or machining operation, and the standard's own guidance is that sinks under large bosses and uneven wall heights are what break it.

FIX Open the flatness to the S-4A-8 value for the surface's largest dimension, specify Precision grade, or machine the face and respect the 0.5 mm skin.

SOURCE NADCA #402, Product Specification Standards for Die Castings, 2021 (11th ed.), Tables S-4A-8-21 and P-4A-8-21 pp.4A-23/24, READ FIRST-HAND: Standard 0.20 mm to 76.2 mm largest dimension + 0.08 mm per additional 25.4 mm; Precision 0.13 + 0.05. Verified against the standard's own worked example - a 254 mm diagonal carries 0.76 mm Standard, 0.48 mm Precision.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

18 Rib taller than the wall can support

LOW

```
maxRibHeightToWall  <=  5 rib height/wall
hpdc-zinc-rib-height
```

A tall rib must fill against die chilling along its whole length, and it also concentrates stress at its base where the section changes.

FIX Reduce the rib height, or use more shorter ribs and generous base fillets.

SOURCE Casting rib-proportion guidance (height against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Low-pressure die casting

11 RULES

01 Wall area below the minimum permanent-mould draft

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
lpdc-draft-minimum  ·  cutoff 1 deg
```

The part shrinks onto a steel mould as it solidifies under pressure. Without taper it grips, and the ejectors either mark the face or tear it.

FIX Add at least 1° of draft to every drawn wall, 2-3° on deep or cosmetic faces.

SOURCE Permanent-mould design guidance: minimum draft about 1°, with 2-3° generally recommended and 2° specified on box-section walls perpendicular to the parting plane. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Local section below what a low-pressure fill reaches

HIGH

```
wallP5Mm  >=  2 mm
lpdc-min-section
```

Low-pressure filling is slow and bottom-fed, which is what makes it clean — and also what limits it. Below about 2 mm the front freezes before the section is full and the result is a misrun or a cold shut, the two defects the process is otherwise chosen to avoid.

FIX Thicken the section to at least 2 mm, or move the part to high-pressure die casting, which fills a 1.0-1.5 mm wall.

SOURCE Low-pressure die casting is quoted at 2.0-3.0 mm thin-wall capability, against gravity die casting requiring 3 mm and up, where cold shuts and misruns appear at 4 mm. Consistent across process-comparison guides; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than low-pressure die casting holds

HIGH

```
tightestToleranceMm  >=  0.6 mm total band
lpdc-tolerance-capability
```

Low-pressure die casting is the most accurate of the permanent-mould routes — ISO 8062 CT6-CT7 against gravity's CT7-CT9 — but it is still a casting. A band tighter than about 0.6 mm total is a machining requirement, and the stock for it has to be on the part.

FIX Open the tolerance, or add machining stock on that feature and quote the operation.

SOURCE Low-pressure die casting quoted at ISO 8062 CT6-CT7 and ±0.3 mm, against gravity die casting at CT7-CT9. Consistent across process-comparison guides; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Non-uniform wall thickness

HIGH

```
wallSpreadRatio  <=  0.7 (p95-p5)/p50
lpdc-wall-uniformity
```

The held pressure feeds shrinkage from the fill tube upward, so a heavy section ABOVE a thin one is cut off from its own feed path and shrinks into porosity. Uniformity matters more here than in gravity casting, not less, because the feed direction is fixed by the process.

FIX Even out the section, or re-orient the part so heavy sections sit low, nearest the fill tube.

SOURCE Casting design guidance on uniform wall thickness and gradual transitions, applied to a bottom-fed directional-solidification process. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Hole too small to be cast — it will be drilled

MEDIUM

```
minHoleDiaMm  >=  6 mm dia
lpdc-min-cored-hole
```

The core hardware is the same steel-in-a-permanent-mould as gravity die casting, and 0.3-1.5 bar is not the 500-1000 bar that lets high-pressure die casting run a finer pin. The permanent-mould floor applies unchanged: 6 mm, parallel to the draw.

FIX Open the hole to at least 6 mm and align it with the draw, or quote it as drilled.

SOURCE Permanent-mould (gravity and low-pressure die) design guidance, citing NADCA: minimum cored hole diameter 6.0 mm, parallel to the direction of draw. The standard is NAMED by the design guide; the NADCA document itself has not been read first-hand.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

06 Cored hole beyond the core slenderness limit

MEDIUM

```
maxHoleDepthToDia  <=  6 core L/D
lpdc-core-ld
```

Past about six diameters a core pin deflects under the metal front and the hole walks off position, taking the wall around it eccentric with it.

FIX Open the diameter, shorten the hole, or support the pin at both ends by taking it through.

SOURCE Die casting core-pin guidance: past an L/D of 6:1 deflection risk rises sharply and additional support, a larger diameter or reduced local pressure is needed. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

07 Cored holes below the core draft

MEDIUM

```
coredHoleDraftPerSideDeg  >=  2 deg per side
lpdc-cored-hole-draft
```

Low-pressure filling is gentler than HPDC but the casting still shrinks onto a steel core, and the pin sees the same binding risk over a longer cycle.

FIX Take cored holes to 2 degrees per side.

SOURCE Permanent-mould and low-pressure die design guidance: cored holes drafted roughly twice the wall figure. Secondary sources, consistent with the wall rule for this family.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

08 Rib heavier than the wall it stands on

MEDIUM

```
maxRibThicknessToWall  <=  1 rib/wall
lpdc-rib-thickness-max
```

A rib thicker than its wall is a heavy section hanging off a thin one. It solidifies last, draws metal it cannot get, and leaves a sink on the show face opposite.

FIX Bring the rib base to 0.6-1.0 times the wall.

SOURCE Die casting rib guidance: rib base thickness 0.6-1.0 × wall, height <= 5 × rib thickness, fillet 1.0-1.25 × rib thickness, draft 1-3°. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

09 Undercuts need a slide or a sand core

MEDIUM

```
undercutFaceCount  <=  0 regions
lpdc-undercuts
```

A permanent mould opens along one axis. An undercut buys either a slide — mechanism, maintenance and cycle time — or a sand core, which reintroduces the sand handling the permanent mould was chosen to escape.

FIX Re-orient the part, or move the undercut feature to a machined operation.

SOURCE Permanent-mould design guidance on parting-line selection and slide cost. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

10 **Rib taller than it can be filled and drawn** LOW

```
maxRibHeightToWall  <=  5 rib height/wall
lpdc-rib-height
```

A tall thin rib is a deep narrow slot in the mould: hard to fill, hard to vent, and hard to draw without tearing.

FIX Keep rib height within five times the wall, or split one tall rib into two shorter ones.

SOURCE Die casting rib guidance: rib height <= 5 × rib thickness. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

11 **Rib too thin to fill** LOW

```
minRibThicknessToWall  >=  0.6 rib/wall
lpdc-rib-thickness-min
```

A rib much thinner than the wall freezes off before the slow low-pressure front reaches its tip, and the stiffness it was drawn for is not there.

FIX Bring the rib base to at least 0.6 of the wall.

SOURCE Die casting rib guidance: rib base thickness 0.6-1.0 × wall. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Gravity die casting

13 RULES

01 **Wall area below the minimum gravity die-casting draft** HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
gdc-draft-minimum  ·  cutoff 1.5 deg
```

The casting solidifies slowly against a steel mould and grips it hard as it contracts, so it needs more draft than a die casting to strip without tearing or scoring the tool.

FIX Allow 1.5-2 degrees on external walls and 2-3 degrees on internal walls and cores.

SOURCE Permanent-mould casting design guidance (draft above die-casting values because of the longer contact time).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 3 copper

02 **Local section below the process minimum** HIGH

```
wallP5Mm  >=  2.5 mm
gdc-min-section
```

Gravity casting fills under its own head. The MEDIAN wall can be healthy while one thin web misruns, and a misrun is scrap rather than a tolerance problem — so the thinnest place is checked separately from the nominal.

FIX Thicken the thinnest region, or move the feature to where the section is already full.

SOURCE Process minimum-section guidance. Checked on the 5th-percentile wall, which is the thinnest measured place rather than the nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 3.5 copper

03 Wall thickness outside the gravity die-casting range

HIGH

```
wallP50Mm  in  3 to 8 mm
gdc-wall-thickness-range
```

Gravity casting fills under its own head, not under pressure, so a thin section misruns before it fills — 3 mm is about the practical floor and 4-6 mm is normal. Because the mould is steel and cools quickly, sections beyond about 8 mm need risering to stay sound. This band is nothing like the die-casting band, and judging a gravity part at 1.0-3.5 mm fails it automatically.

FIX Hold 4-6 mm as the nominal wall. If the design needs a section thinner than 3 mm, gravity casting is the wrong process — look at high-pressure die casting.

SOURCE Gravity / permanent-mould aluminium casting design guidance: 3 mm practical minimum, 4-6 mm typical.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

4-10 copper **3-8** aluminium

04 Non-uniform wall thickness

HIGH

```
wallSpreadRatio  <=  0.7 (p95-p5)/p50
gdc-wall-uniformity
```

Gravity casting relies on directional solidification: metal must freeze progressively towards a riser. An isolated heavy section freezes last with no feed path and draws a shrinkage cavity, which is the dominant defect in this process.

FIX Taper the section towards the risers so the casting freezes progressively, and core out isolated heavy masses.

SOURCE Permanent-mould casting design guidance (directional solidification and riser feeding).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 0.6 copper

05 Tolerance tighter than gravity die casting holds

HIGH

```
tightestToleranceMm  >=  0.6 mm total band
gdc-tolerance-capability
```

A gravity casting cools slowly against a steel mould and moves as it does. About ISO 8062 CT8-10 is normal, which is roughly +/-0.3 mm on a 50 mm dimension.

FIX Open the as-cast tolerance and machine the functional faces.

SOURCE ISO 8062 casting tolerance grades CT8-CT10 (permanent mould).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Hole too small to be cast — it will be drilled

MEDIUM

```
minHoleDiaMm  >=  6 mm dia
gdc-min-cored-hole
```

A gravity-poured permanent mould has no shot pressure to force metal down a fine pin, and the pin itself sits in a hot die for a long, slow fill. The published permanent-mould floor is 6 mm, and the hole must run parallel to the draw. Anything finer is a drilled feature wearing a cast feature's drawing callout.

FIX Open the hole to at least 6 mm and align it with the draw direction, or delete it from the casting and quote the drilling.

SOURCE Permanent-mould (gravity die) design guidance, citing NADCA: minimum cored hole diameter 6.0 mm, parallel to the direction of draw. The standard is NAMED by the design guide; the NADCA document itself has not been read first-hand.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

07 Boss taller than the core can hold

MEDIUM

```
maxBossHeightToDia  <=  3 height/dia
gdc-boss-height
```

A gravity-cast boss is cored by steel or bonded sand and neither is injected against, but a slender core still floats and distorts as the metal rises around it.

FIX Shorten the boss, widen it, or cast it solid and drill afterwards.

SOURCE Permanent-mould boss-proportion guidance.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

08 Cored hole beyond the core slenderness limit

MEDIUM

```
maxHoleDepthToDia  <=  6 core L/D
gdc-core-ld
```

Gravity die cores are steel or bonded sand and are not injected against, but a slender core still floats, distorts or breaks on extraction, and a sand core has no strength to spare.

FIX Shorten the cored hole, open its diameter, or drill it after casting.

SOURCE Permanent-mould casting design guidance (core slenderness limits).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

09 Cored holes below the core draft

MEDIUM

```
coredHoleDraftPerSideDeg  >=  3 deg per side
gdc-cored-hole-draft
```

Permanent-mould cores are steel and the casting shrinks onto them through a longer, slower solidification than die casting, so cored holes need more draft than the 1.5 deg the walls take.

FIX Take cored holes to 3 degrees per side, or use a sand core where the hole must stay parallel.

SOURCE Permanent-mould (gravity die) design guidance: cored holes drafted roughly twice the wall figure. Secondary sources, consistent with the same guidance the wall rule cites.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

10 Rib heavier than the wall it stands on

MEDIUM

```
maxRibThicknessToWall  <=  0.8 rib/wall
gdc-rib-thickness-max
```

A gravity casting freezes slowly, so a rib fuller than about 80% of the wall becomes a hot spot with no feed path and draws shrinkage into the junction.

FIX Thin the rib, or add two thinner ribs instead of one heavy one.

SOURCE Casting rib-proportion guidance (rib thickness against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

11 Rib too thin to fill before it freezes

MEDIUM

```
minRibThicknessToWall  >=  0.5 rib/wall
gdc-rib-thickness-min
```

A rib is the last thing to fill and the first thing to freeze, because it presents the most surface for its volume. Below this fraction of the nominal wall it will not fill reliably.

FIX Thicken the rib towards the process minimum, or shorten it so the metal has less distance to travel.

SOURCE Casting rib-proportion guidance (fill limit against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

12 **Undercuts complicate the tool** MEDIUM

```
undercutFaceCount  <=  0 regions
gdc-undercuts
```

A permanent mould opens on a fixed axis. An undercut needs a loose piece or a sand core set into the steel die, which adds a core-making operation and a setting step to every cycle.

FIX Re-orient the part on the mould split, or design the feature to be formed by the sand core that is already there.

SOURCE Process tooling guidance (split lines, loose pieces and cores).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

13 **Rib taller than the wall can support** LOW

```
maxRibHeightToWall  <=  4 rib height/wall
gdc-rib-height
```

A tall rib must fill against die chilling along its whole length, and it also concentrates stress at its base where the section changes.

FIX Reduce the rib height, or use more shorter ribs and generous base fillets.

SOURCE Casting rib-proportion guidance (height against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Squeeze casting

6 RULES

01 **Wall too thin for a slow-filled squeeze casting** HIGH

```
wallP5Mm  >=  3 mm
squeeze-min-section
```

Squeeze casting fills the cavity in one to three seconds to keep the metal free of turbulence and gas, then holds it under about 100 MPa while it freezes. That slow fill is what makes the casting dense enough to heat treat — and it is also why a thin wall solidifies before the metal gets there.

FIX Take the thinnest section to at least 3 mm, or choose conventional or vacuum die casting, which fill fast enough for a thinner wall.

SOURCE NADCA Product Design for Die Casting, 7th ed. (2015), §5.2 p.129, READ FIRST-HAND: "As the castings are filled so slowly, components produced by squeeze casting are normally thicker walled (minimum of 0.12-0.20 inches/3-5 mm) so that the liquid aluminum doesn't solidify before the casting is fully filled." The lower end of the published band is used here; 5 mm is the typical figure, not the limit.

NAMED STANDARD, NOT READ FIRST-HAND · NOT YET AUDITED

02 **Wall area below the minimum squeeze-casting draft** HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
squeeze-draft-minimum  ·  cutoff 1.5 deg
```

The part is held against the die under load while it solidifies, so it grips harder than a free-shrinking casting. Zero-draft walls that a die caster would accept will not release here.

FIX Add at least 1.5° of draft to every drawn wall, 2-3° on deep pockets.

SOURCE Die casting draft guidance of 1-3°, with 2-3° minimum on deep pockets, applied to a process that solidifies under applied load. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than squeeze casting holds

HIGH

```
tightestToleranceMm  >=  0.6 mm total band
squeeze-tolerance-capability
```

Squeeze casting works in a steel die and holds a permanent-mould class of tolerance. A band tighter than about 0.6 mm total is a machining requirement.

FIX Open the tolerance, or leave machining stock and quote the operation.

SOURCE Permanent-mould tolerance class (ISO 8062 CT6-CT8) applied to a steel-die process. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Cored hole beyond the core slenderness limit

MEDIUM

```
maxHoleDepthToDia  <=  5 core L/D
squeeze-core-ld
```

A core pin in a squeeze die is loaded by the ram for the whole solidification, not by a shot for milliseconds. The slenderness it survives is below the die-casting figure, not above it.

FIX Open the diameter, shorten the hole, or drill it afterwards.

SOURCE Die casting core-pin guidance (deflection risk rising sharply past L/D 6:1), tightened for a core loaded through solidification rather than during fill. Derived, not quoted — treat as a screening value.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Undercuts need a slide in a pressurised die

MEDIUM

```
undercutFaceCount  <=  0 regions
squeeze-undercuts
```

A slide in a squeeze die carries the full press load, not just a shot pressure spike. It is a more expensive mechanism than the same slide in a gravity mould and it is the first thing to wear.

FIX Re-orient the part, or machine the undercut afterwards.

SOURCE Permanent-mould design guidance on parting-line selection and slide cost, applied to a load-bearing die. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Non-uniform wall thickness

MEDIUM

```
wallSpreadRatio  <=  0.9 (p95-p5)/p50
squeeze-wall-uniformity
```

Applied pressure feeds shrinkage, which is precisely why squeeze casting tolerates a heavier section than gravity does — but the pressure has to reach it. A heavy pocket isolated behind a thin wall freezes off from the feed path and shrinks anyway.

FIX Even out the section, or place the heavy section where the ram pressure acts directly on it.

SOURCE Casting design guidance on uniform wall thickness, relaxed for a process whose applied pressure feeds solidification shrinkage. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Semi-solid casting (thixo / rheo)

6 RULES

01 Local section below what a semi-solid slurry fills

HIGH

```
wallP5Mm  >=  0.5 mm
ssm-min-section
```

A semi-solid slurry fills thinner than liquid metal because it does not spray and freeze off at the front. Walls down to 0.5 mm over palm-sized areas are documented — but 0.5 mm is the floor, not the target.

FIX Thicken the section to at least 0.5 mm; anything below that is not a casting feature.

SOURCE Thixomolding design guidance: wall thickness as low as 0.5 mm over 150 × 100 mm areas, with parts up to 18-19 mm thick also moulded. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Tolerance tighter than semi-solid casting holds

HIGH

```
tightestToleranceMm  >=  0.4 mm total band
ssm-tolerance-capability
```

Low shrinkage and a laminar fill make semi-solid the most repeatable of the die routes, so it holds a tighter band than plain HPDC. It is still a die process with thermal growth in the tool, and a band under about 0.4 mm total belongs to machining.

FIX Open the tolerance, or leave stock and machine the feature.

SOURCE Semi-solid processing is quoted as holding tighter dimensional repeatability than conventional die casting on account of reduced solidification shrinkage. Positioned tighter than the HPDC band used elsewhere in this catalogue; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Wall area below the minimum semi-solid draft

MEDIUM

```
wallAreaBelowDraftPct  <=  10 % of wall area
ssm-draft-minimum  ·  cutoff 0.5 deg
```

This is the one die process that draws with almost no taper — zero draft has been used in production. The rule is here to catch a wall with NO taper across a large area, not to demand the 1-2° a die caster would ask for. A part flagged by the HPDC draft rule is very often fine here, and that difference is the reason to choose the route.

FIX Add half a degree where the drawn area is large, or confirm zero-draft ejection with the moulder before committing.

SOURCE Thixomolding design guidance: parts can be moulded with minimum draft angles or none at all, and zero draft has been used. The threshold here is deliberately permissive against the 1-3° die-casting norm. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Rib heavier than the wall it stands on

MEDIUM

```
maxRibThicknessToWall  <=  1 rib/wall
ssm-rib-thickness-max
```

A rib heavier than its wall is a late-freezing section on an early-freezing one, and it sinks the show face opposite whatever the fill physics.

FIX Match the rib base to the wall or bring it below it.

SOURCE Thixomolding design guidance: match bosses and studs to wall thickness. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Undercuts need a slide

MEDIUM

```
undercutFaceCount  <=  0 regions
ssm-undercuts
```

Low draft does not mean no draw. The die still opens along one axis, and an undercut still buys a slide or a lifter.

FIX Re-orient the part on the parting line, or machine the undercut afterwards.

SOURCE Thixomolding design guidance on gating, ejection and tool opening. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Non-uniform wall thickness

MEDIUM

```
wallSpreadRatio  <=  0.9 (p95-p5)/p50
ssm-wall-uniformity
```

Laminar filling and a low superheat mean less solidification shrinkage than a fully liquid shot, so the section change this process tolerates is wider than HPDC's. It is not unlimited: a heavy boss on a thin wall still sinks.

FIX Even out the section, or core out the heavy region.

SOURCE Thixomolding design guidance: concentrate on uniform wall thickness and cavity filling; match bosses and studs to wall thickness. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Sand casting

15 RULES

01 Wall area below the minimum sand-casting draft

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
sand-draft-minimum  ·  cutoff 1.5 deg
```

The pattern is drawn out of rammed sand. Without taper it tears the mould wall on the way out, and the casting carries the damage. Sand needs the most draft of any casting process.

FIX Allow 1.5-3 degrees on all drawn surfaces, and more on deep pockets.

SOURCE Sand-casting design guidance (pattern draw taper, typically 1.5-3 degrees).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 3 ferrous   **<= 7** castiron

02 Local section below the process minimum

HIGH

```
wallP5Mm  >=  3.5 mm
sand-min-section
```

A sand mould is weak and the metal must carry itself through it. One thin web below about 3.5 mm in aluminium misruns or erodes the sand into the casting, whatever the median section says.

FIX Thicken the thinnest region, or move the feature to where the section is already full.

SOURCE Process minimum-section guidance. Checked on the 5th-percentile wall, which is the thinnest measured place rather than the nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · PRIMARY SOURCE READ

BY MATERIAL FAMILY

>= 6 ferrous   **>= 4** castiron   **>= 4** copper

03 Tolerance tighter than sand casting holds

HIGH

```
sfsaSandToleranceMargin  >=  1 x SFSA 2000 capability
sand-tolerance-capability
```

Sand moves. The mould is rammed, the cores are set by hand and the casting contracts against a yielding medium. For steel the capability is now COMPUTED from the SFSA 2000 CT tables at the feature's own dimension and the declared production series; other alloys are screened at the process band, and the finding says which happened.

FIX Machine every functional surface and open everything else — or negotiate a tighter CT grade with the foundry and pay for the pattern re-engineering it implies. A tight as-cast tolerance on a sand casting is a scrap rate, not a specification.

SOURCE Process screening value (1.2 mm total band) for alloys the SFSA tables do not cover; ISO 8062 grade selection for non-ferrous sand alloys is not held.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · PRIMARY SOURCE READ

BY MATERIAL FAMILY

>= 1 ferrous   **>= 1** castiron

04 Wall thickness outside the sand-casting range

HIGH

```
wallP50Mm  in  4 to 12 mm
sand-wall-thickness-range
```

Sand insulates, so the metal stays liquid longer and fills a long path — but the mould is weak and the section must carry itself, and below about 4 mm in aluminium (5 mm in iron) the casting misruns or the sand erodes. Because cooling is slow, sand tolerates far heavier sections than any die process.

FIX Hold 5-8 mm as the nominal wall. A part needing a 2 mm wall is not a sand casting.

SOURCE Sand-casting design guidance: about 4 mm minimum in aluminium and 5 mm in grey iron; 5-10 mm typical nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

5-15 castiron
6-18 ferrous
4-12 aluminium
5-12 copper

05 Junction fillet below the steel-casting band

MEDIUM

```
sfsaJunctionFilletMargin  >=  1 x Fig. 11 fillet
sand-junction-fillet
```

A steel T-junction wants a fillet equal to the thinner joining wall — but clamped between 13 and 25 mm, because past an inch the fillet GROWS the inscribed circle at the joint and the mass increase goes as (D/d)^2. A sharp junction tears on solidification; an oversized one feeds a shrinkage cavity.

FIX Open the junction fillet to the thinner wall thickness, within the 13-25 mm band; past a 1.5x wall difference keep r at the thinner wall and bridge with a 15 deg slope.

SOURCE SFSA Steel Castings Handbook Supplement 1, Fig. 11 (p.6), READ FIRST-HAND: "r = T1, but never less than 1/2 in. (13 mm) or greater than 1 in. (25 mm)"; the slope past T2 = 1.5 T1 must run at least T2 - T1. Judged against the part's smallest internal corner radius. Steel castings only.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

06 Machining stock below the required machining allowance

MEDIUM

```
sfsaMachiningStockMargin  >=  1 x Table 2.2 RMA
sand-machining-stock
```

The required machining allowance is what guarantees a machined surface CLEANS UP: the as-cast surface sits somewhere inside the casting tolerance, and the cutter must reach fresh metal below the worst of it. Stock below the RMA risks a surface that still carries as-cast skin after machining.

FIX Increase the machining stock to the Table 2.2 allowance for the casting's size — and note the printed 6 mm minimum on cope surfaces, which is the foundry's orientation choice.

SOURCE SFSA Supplement 3, Table 2.2 (p.5), READ FIRST-HAND: required machining allowance by largest casting dimension, grades E-K; machine-molded sand uses F-H and is judged here at grade F, the least the standard ever requires. Steel castings only; abstains when no stock is declared.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

07 Steel rib taller than its thermally neutral height

MEDIUM

```
sfsaRibNeutralityMargin  >=  1 x Fig. 8 neutral height
sand-rib-thermal-neutrality
```

For steel the allowed rib height is COUPLED to rib thickness: SFSA prints neutrality to 4 walls of height only for a rib of 1/4 wall; at 1/2 wall the limit is 1.5 walls, at 3/4 wall just 0.5. A single flat height ceiling passes exactly the ribs — full-thickness, tall — that pull shrinkage into the joint. Too thin fails the other way: the rib acts as a cooling fin and upsets the freezing order.

FIX Thin the rib to move up the neutrality curve, shorten it to its own neutral height, or split it into staggered ribs at least 7 rib-thicknesses apart.

SOURCE SFSA Steel Castings Handbook Supplement 1, Fig. 8 (p.5), READ FIRST-HAND: T1 = 1/4 T2 thermally neutral until h = 4 T2; 1/2 until 1.5 T2; 3/4 until 0.5 T2; spacing at least 7 T1. Interpolated at each rib's own ratio; the worst rib on the part sets the margin and is named. Steel castings only – the measure abstains on every other alloy.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

08 Rib heavier than the wall it stands on

MEDIUM

```
maxRibThicknessToWall  <=  1 rib/wall
sand-rib-thickness-max
```

Sand cools slowest of all, so it tolerates the fullest ribs — but a rib heavier than the wall it stands on inverts the freezing order and pulls a shrinkage cavity into the joint.

FIX Thin the rib, or add two thinner ribs instead of one heavy one.

SOURCE Casting rib-proportion guidance (rib thickness against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · PRIMARY SOURCE READ

BY MATERIAL FAMILY

<= 0.75 ferrous

09 Cored hole beyond the sand-core slenderness limit

MEDIUM

```
maxHoleDepthToDia  <=  4 core L/D
sand-core-ld
```

A bonded sand core has almost no strength in bending and is buoyant in liquid metal. A slender core floats, sags or washes away, and the hole ends up bent or absent.

FIX Open the hole, support the core on prints at both ends, or drill after casting.

SOURCE Sand-casting design guidance (core strength and buoyancy).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

10 Rib too thin to fill before it freezes

MEDIUM

```
minRibThicknessToWall  >=  0.6 rib/wall
sand-rib-thickness-min
```

A rib is the last thing to fill and the first thing to freeze, because it presents the most surface for its volume. Below this fraction of the nominal wall it will not fill reliably.

FIX Thicken the rib towards the process minimum, or shorten it so the metal has less distance to travel.

SOURCE Casting rib-proportion guidance (fill limit against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

11 Non-uniform wall thickness

MEDIUM

```
wallSpreadRatio  <=  0.9 (p95-p5)/p50
sand-wall-uniformity
```

Slow cooling in sand gives a heavy section time to feed from a riser, so sand tolerates more section change than any die process — but an isolated hot spot with no feed path still draws shrinkage.

FIX Taper sections towards the risers and avoid isolated heavy bosses far from a feed path.

SOURCE Sand-casting design guidance (feeding distance and hot spots).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

12 Boss more than twice the wall it stands on

LOW

```
sfsaBossDiaToWall  <=  2 boss dia / wall
sand-boss-dia
```

SFSA's boss-neutrality bands stop at a boss of twice the parent wall — and even that is neutral only to a quarter-wall of height. Beyond 2 T no neutral height is printed at all: the boss is an isolated mass that must be fed or it shrinks, and bosses are the textbook place centre-line shrinkage hides.

FIX Shrink the boss toward the wall thickness, blend it with tapered fillets, join several bosses into one machined panel, or core the centre out of a heavy one.

SOURCE SFSA Steel Castings Handbook Supplement 1, Fig. 16 + Table 1 (p.7), READ FIRST-HAND: neutrality bands D = 1.0/1.3/2.0 T with heights 1.5/1.0/0.25 T (less with fillets); spacing 3 T minimum. The worst boss on the part sets the ratio. Steel castings only.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

13 Flatness callout tighter than sand casting holds

LOW

```
sfsaSandFlatnessMargin  >=  1 x CTG7 flatness
sand-flatness-capability
```

A sand casting's faces move with the mould: ISO 8062-2 puts machine-molded steel sand casting at CTG5-7, and flatness capability grows with the length of the face carrying it. A callout tighter than the CTG band is a grinding or machining operation, not an as-cast property.

FIX Open the flatness callout to the CTG capability, or plan to machine or gage-grind the surface that carries it.

SOURCE SFSA Supplement 3, Tables 4.2a + 4.5 (ISO 8062-2 based), READ FIRST-HAND: flatness by casting geometrical tolerance grade; steel sand castings machine-molded CTG5-7 (hand molding 6-8). Judged at CTG7 at the part's bounding diagonal. Steel castings only — the measure abstains on other alloys, and with no declared callout.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

14 Rib taller than the wall can support

LOW

```
maxRibHeightToWall  <=  4 rib height/wall
sand-rib-height
```

A tall rib must fill against die chilling along its whole length, and it also concentrates stress at its base where the section changes.

FIX Reduce the rib height, or use more shorter ribs and generous base fillets.

SOURCE Casting rib-proportion guidance (height against nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

15 Undercuts require a core or a loose pattern piece

LOW

```
undercutFaceCount  <=  0 regions
sand-undercuts
```

Sand casting handles undercuts more cheaply than any die process — a core box or a loose pattern piece, not a hydraulic slide in hardened steel. Each one still adds a core to make, set and remove, so the count is a cost driver rather than a feasibility limit.

FIX Where a core is unavoidable, design it to be set on prints at both ends so it cannot float.

SOURCE Sand-casting design guidance (cores and loose pieces).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Shell mould casting

5 RULES

01 Wall area below the minimum shell-mould draft

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
shell-draft-minimum  ·  cutoff 1 deg
```

The pattern is drawn from a cured resin shell, which is rigid — so it needs less taper than green sand, where the mould crumbles if the pattern rubs, but more than nothing.

FIX Add at least 1° of draft to every drawn wall.

SOURCE Casting draft allowance guidance of 1-3° per vertical face, taken at the lower end because the cured shell is rigid where green sand is not. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Local section below what a shell mould fills

HIGH

```
wallP5Mm  >=  2.5 mm
shell-min-section
```

A resin shell chills faster than rammed green sand, so it fills a thinner section than green sand does — but it is still a gravity pour into a sand-class mould, not a pressurised one.

FIX Thicken the section, or move the part to a permanent-mould or pressurised route.

SOURCE Shell moulding is quoted as reaching thinner sections and a better finish (Ra 1.6-3.2 µm) than green sand casting. The figure here sits between the sand-casting minimum used elsewhere in this catalogue and the permanent-mould minimum; it is DERIVED from that ordering, not quoted, and should be the first threshold a foundry review corrects.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than shell moulding holds

HIGH

```
tightestToleranceMm  >=  0.8 mm total band
shell-tolerance-capability
```

A rigid cured shell holds a tighter band than rammed green sand — that accuracy, with the finish, is the whole reason to pay for the heated pattern — but it does not approach a steel die.

FIX Open the tolerance, or leave machining stock and quote the operation.

SOURCE Shell moulding sits between green sand and permanent mould on the ISO 8062 CT scale. The value here is positioned between the sand-casting and gravity-die bands used elsewhere in this catalogue; DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Non-uniform wall thickness

HIGH

```
wallSpreadRatio  <=  1 (p95-p5)/p50
shell-wall-uniformity
```

A gravity pour feeds shrinkage through risers, and a riser only feeds a section it can still reach through liquid metal. Isolated heavy sections shrink whatever the mould is made of.

FIX Even out the section, or add a feeder path to the heavy region.

SOURCE Casting design guidance on uniform wall thickness and gradual transitions. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Cored hole beyond the sand-core slenderness limit

MEDIUM

```
maxHoleDepthToDia  <=  10 core L/D
shell-core-ld
```

A resin-bonded core is far stronger than a green-sand one, but it is still bonded sand. Past roughly ten diameters unsupported it is too brittle to handle without damage, let alone to survive the pour.

FIX Support the core at both ends, open the diameter, or drill the hole after casting.

SOURCE Foundry core guidance: an unsupported core length-to-diameter ratio beyond about 10-15 forces alternatives to conventional sand cores. The lower end of that band is used here. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Investment casting

11 RULES

01 Local section below the process minimum

HIGH

```
wallP5Mm  >=  1.2 mm
inv-min-section
```

The preheated shell keeps metal fluid a long way, but below about 1.2 mm even investment casting misruns. This is the process floor, not the design target.

FIX Thicken the thinnest region, or move the feature to where the section is already full.

SOURCE Process minimum-section guidance. Checked on the 5th-percentile wall, which is the thinnest measured place rather than the nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 1.6 ferrous   **>= 2** titanium

02 Tolerance tighter than investment casting holds

HIGH

```
sfsaInvToleranceMargin  >=  1 x CT5-7 capability
inv-tolerance-capability
```

The wax pattern is moulded to a machined die and the ceramic shell is rigid, so investment casting holds ISO 8062 CT5-7 — the accuracy the process is bought for. For steel the capability is COMPUTED from the CT table at the feature's own dimension; other alloys are screened at a 0.3 mm total band, and the finding says which happened.

FIX Hold the tolerance as-cast where you can; if it must be tighter, machine only that feature.

SOURCE Process screening value (0.3 mm total band, roughly CT5-7 at small dimensions) for alloys the SFSA tables do not cover.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · PRIMARY SOURCE READ

BY MATERIAL FAMILY

>= 1 ferrous

03 Wall thickness outside the investment-casting range

HIGH

```
wallP50Mm  in  1.5 to 6 mm
inv-wall-thickness-range
```

The ceramic shell is preheated before pouring, so the metal stays fluid and fills thinner than sand allows — about 1.5 mm routinely and under 1 mm on small parts. Heavy sections are limited by feeding, as in any gravity process.

FIX Hold 2-4 mm as the nominal wall; investment casting is chosen for thin, complex sections, so leaving a heavy wall wastes the process.

SOURCE Investment-casting design guidance: about 1.5 mm general minimum, 2-4 mm typical.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

2-6 ferrous   **2.5-8** titanium   **1.5-5** aluminium

04 Blind cored hole too deep for an unsupported ceramic core

HIGH

```
maxBlindHoleDepthToDia  <=  2 blind core L/D
inv-blind-core-ld
```

A blind hole is formed by a ceramic core held at one end only. Past about two diameters the cantilever cannot survive wax injection and shell build, and it snaps off inside the pattern where nobody sees it until the casting is sectioned.

FIX Reduce the depth to twice the diameter, open the diameter, or take the hole through so the core can be supported at both ends.

SOURCE Investment casting design guidance: depth/diameter ratio for BLIND holes should be under 2, and for THROUGH holes under 5. Consistent across several foundry design guides; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Cored hole beyond the ceramic-core slenderness limit

HIGH

```
maxHoleDepthToDia  <=  4 core L/D
inv-core-ld
```

An investment-cast internal passage is made by a ceramic core that has to survive wax injection, shell building, burnout and pouring, and then be leached out. Ceramic cores are the most fragile of any casting process.

FIX Open the passage, shorten it, or accept a drilled hole after casting.

SOURCE Investment-casting design guidance (ceramic core fragility).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Non-uniform wall thickness

MEDIUM

```
wallSpreadRatio  <=  0.7 (p95-p5)/p50
inv-wall-uniformity
```

The shell is a poor conductor, so an isolated heavy section stays liquid after its feed path has frozen and draws a shrinkage cavity.

FIX Blend section changes and place heavy masses where a feeder can reach them.

SOURCE Investment-casting design guidance (feeding and hot spots).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 0.6 ferrous

07 Hole too small for a ceramic core

MEDIUM

```
minHoleDiaMm  >=  1.5 mm dia
inv-min-cored-hole
```

Investment casting makes a hole with a ceramic core inside a wax pattern, not with a steel pin. Ceramic is brittle but it is also formed, not machined, so it goes finer than any die pin — down to about 1.5 mm. Below that the core breaks during wax injection or dewax and the hole closes.

FIX Open the hole to at least 1.5 mm, or drill it after casting — investment castings machine easily and the stock is already there.

SOURCE Investment casting design guidance: minimum hole diameter approximately 1.5 mm (some houses quote 2.0 mm for through holes), with tolerances of the order of ±0.13 mm up to 25 mm. Consistent across several foundry design guides; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

08 Through cored hole too deep for a ceramic core

MEDIUM

```
maxThroughHoleDepthToDia  <=  5 through core L/D
inv-through-core-ld
```

A through hole lets the ceramic core be anchored at both ends, which roughly doubles the slenderness it survives against a blind one. It is still ceramic: past about five diameters it bows during shell build and the bore goes out of straightness.

FIX Open the diameter, shorten the hole, or accept a drilled bore and leave machining stock.

SOURCE Investment casting design guidance: depth/diameter ratio for THROUGH holes should be under 5, against under 2 for blind holes. Consistent across several foundry design guides; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

09 Flatness callout tighter than investment casting holds

LOW

```
sfsaInvFlatnessMargin  >=  1 x CTG6 flatness
inv-flatness-capability
```

Investment casting holds the tightest geometric grades of any casting route — ISO 8062-2 CTG4-6 for steel — but flatness still grows with the face length, and a callout tighter than the band is a finishing operation.

FIX Open the flatness callout to the CTG capability, or machine the one surface that needs it.

SOURCE SFSA Supplement 3, Tables 4.2a + 4.5 (ISO 8062-2 based), READ FIRST-HAND: steel investment castings CTG4-6, judged at CTG6 at the part's bounding diagonal. Steel castings only; abstains with no declared callout.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

10 Wall area below the minimum investment-casting draft

LOW

```
wallAreaBelowDraftPct  <=  10 % of wall area
inv-draft-minimum  ·  cutoff 0.5 deg
```

Nothing is drawn out of the ceramic shell — it is broken away. Draft is needed only to release the wax pattern from its own die, and zero-draft walls are routinely cast. This is the reason to choose investment casting for a part that cannot carry draft.

FIX Allow a nominal 0.5 degrees where it costs nothing; genuinely zero-draft walls are acceptable and should be discussed with the founder rather than redesigned.

SOURCE Investment-casting design guidance (draft needed only for wax-pattern release).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

11 Undercuts complicate the tool

LOW

```
undercutFaceCount  <=  0 regions
inv-undercuts
```

The ceramic shell is broken away, so the CASTING has no undercut problem — but the wax pattern still has to leave its own die, and that die is a two-part tool. An undercut moves the cost into the wax tooling rather than removing it.

FIX Re-orient the pattern on its die split, or accept a loose piece in the wax tool.

SOURCE Process tooling guidance (split lines, loose pieces and cores).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Centrifugal casting

3 RULES

01 Part is not a body of revolution

HIGH

```
axisymmetricAreaPct  >=  90 % of surface area
cent-body-of-revolution  ·  BLOCKING
```

Centrifugal casting forms the part by spinning the mould: the outer surface is the mould bore and the inner surface is a free liquid surface held by rotation. A feature that is not axisymmetric — a lug, a flat, a bolt boss — cannot be produced by that motion at all. This is not a tolerance to negotiate; it is whether the route exists for this part.

FIX Cast an axisymmetric blank and machine the non-axisymmetric features, or choose a route with a static mould.

SOURCE Derived from the process kinematics rather than from a design guide: a spinning mould can only generate surfaces of revolution about its own axis. The 90% threshold allows for small machined-in features on an otherwise round part; it is this tool's own screening value, not a published one.

THIS TOOL'S OWN COST MODEL · NOT YET AUDITED

02 Tolerance tighter than centrifugal casting holds

HIGH

```
tightestToleranceMm  >=  0.5 mm total band
cent-tolerance-capability
```

Rotation gives a dense, well-controlled OUTER diameter — the tightest as-cast surface of any sand-class process. The bore is a free liquid surface and is not controlled at all; it is machined.

FIX Open the tolerance, or apply the tight band to the outer diameter only and machine the bore.

SOURCE Centrifugal casting is quoted at radial tolerance CT3-CT8 (±0.1-0.5%) with a surface finish of Ra 3-8 µm. The value here takes the loose end of that band as a screening figure; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Local section below what a spun pour fills

MEDIUM

```
wallP5Mm  >=  5 mm
cent-min-section
```

Centrifugal castings are liners, rings and bushings: the wall is set by the mould bore and the pour volume, and the inner surface carries the dross that has to be machined off. A thin as-cast wall leaves nothing to remove it from.

FIX Thicken the as-cast wall, or choose a route that casts closer to net shape.

SOURCE Centrifugal casting practice: the bore is machined to remove the dross layer that collects on the inside diameter, so the as-cast wall must carry that allowance on top of the finished wall. Machining allowances of 1-3 mm are quoted for casting processes generally. DERIVED from that practice, not quoted as a wall minimum.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

CHAPTER 3

Sheet metal and forming

Flat stock cut and bent. Nearly every limit here is a ratio to material thickness, and most are material-dependent because the alloy decides how far it will stretch before it splits.

42 RULES · 7 PROCESSES · 23 HIGH / 19 MEDIUM / 0 LOW · 12 CARRY MATERIAL-SPECIFIC THRESHOLDS

Sheet metal / stamping

12 RULES

01 Strip utilisation below the industry target

HIGH

`blankUtilisationPct >= 70 % of strip area`

`sm-blank-utilisation`

The book opens its material-economy chapter with 'the major portion of the cost of producing a stamped component is the material' and sets the target at 70 to 80 percent of the strip. On a stamping this is usually the largest single cost lever on the part, and it is invisible on a report that quotes only a piece price.

FIX Re-lay the blank: rotate it, nest it two-up or interlocked, or alter the outline where function allows. The book devotes Sec. 4.4.2 to changing the workpiece design for exactly this.

SOURCE Boljanovic, Sec. 4.4 with Eq. 4.7 and Table 4.3 for the webs; the 70-80% target is stated in the text. READ FIRST-HAND.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

02 Tolerance tighter than sheet metal / stamping holds

HIGH

`tightestToleranceMm >= 0.4 mm total band`

`sm-tolerance-capability`

A formed feature carries springback, and springback varies with the coil, not just the tool. About +/-0.2 mm on a formed dimension is normal; a pierced hole in the flat is far better than a formed feature and should be toleranced separately.

FIX Tolerance pierced features tightly and formed features loosely, and datum from the pierced ones.

SOURCE Sheet-metal forming tolerance guidance (springback-dominated, formed vs pierced features).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 0.8	Steel DP980 (dual-phase)	>= 0.3	Steel 22MnB5 (press-hardened)	>= 0.4	Aluminium 5052 (sheet)
>= 0.6	Steel (high-strength)	>= 0.6	Aluminium 6061		

03 Hole too close to a bend line

HIGH

`holeToBendClearanceMm >= 0 mm clear of 2t+r`

`sm-hole-to-bend`

A hole inside the bend deformation zone is pulled oval as the bend forms, and no amount of press setup recovers the shape.

FIX Move the hole clear of the bend zone (at least 2 thicknesses plus the bend radius), or pierce it after forming.

SOURCE Sheet-metal design guidance (hole-to-bend >= 2T + R).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Inside bend radius tighter than the grade will take

MEDIUM

```
minBendRadiusToThickness  >=  1 r/t
sm-bend-radius
```

Bending tighter than about one thickness works the outer fibre beyond its uniform elongation and cracks it. Less ductile alloys need considerably more.

FIX Open the inside radius to at least the r/t shown above for this grade — the figure is the alloy's, not a generic one. Where the radius has to stay, a hot or annealed bend is the alternative.

SOURCE Sheet-metal design guidance (1x t typical, 1.5–2x for stainless and 6061-T6).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · PRIMARY SOURCE READ

BY MATERIAL GRADE

>= **2.5** Steel DP600 (dual-phase)	>= **4** Steel DP980 (dual-phase)	>= **6** Steel 22MnB5 (press-hardened)
>= **1** Stainless Steel 316L	>= **1.5** Stainless Steel 430	>= **1** Aluminium 5052 (sheet)
>= **3** Aluminium 6082	>= **3** Aluminium 6061	>= **4** Aluminium 7075
>= **2** Steel (high-strength)	>= **1** Stainless Steel 304	>= **1** Steel (mild)

05 Bend radius too LARGE to take a permanent set

MEDIUM

```
bendRadiusMaxRatio  <=  1 x the maximum that will yield
sm-bend-radius-max
```

Every DFM tool checks that a bend is not too tight. A bend that is too GENTLE never strains the outer fibre past yield, so on unloading the part springs back to flat and the feature does not exist. The limit is T*E/(2*YS), which on thin high-yield sheet is reached at surprisingly modest radii.

FIX Tighten the radius below T*E/(2*YS), or form the feature by another means — a large-radius contour is a stretch-forming or roll-forming job, not a brake bend.

SOURCE Boljanovic, 'Sheet Metal Forming Processes and Die Design', Eq. 5.15: R_i(max) <= T*E/(2*YS). READ FIRST-HAND.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

06 Parallel bends too close to form

MEDIUM

```
minBendToBendToThickness  >=  4 flat/t
sm-bend-to-bend
```

A press brake needs flat material to clamp between two bends. When the land between parallel bends falls below roughly four thicknesses there is nothing for the tooling to hold, and the second bend pulls the first out of angle.

FIX Open the land between the bends to at least 4t, or form them in one operation with a dedicated tool.

SOURCE Sheet-metal design guidance (minimum flat between parallel bends).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= **6** Aluminium 6061	>= **5** Steel (high-strength)

07 Flange too short to form

MEDIUM

```
minFlangeToThickness  >=  3 flange/t
sm-flange-length
```

A flange shorter than about three thicknesses gives the press-brake tooling nothing to hold, so the bend wanders and the angle is not repeatable.

FIX Lengthen the flange to at least 3x thickness, or form it as part of a larger feature and trim after.

SOURCE Sheet-metal design guidance (minimum flange ~3x thickness).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= **4** Aluminium 6061	>= **4** Steel (high-strength)

08 Hole too close to the part edge

MEDIUM

```
minHoleToEdgeToThickness  >=  2 gap/t
sm-hole-to-edge
```

A hole pierced near a trimmed edge pushes material sideways with nothing to react against, so the edge bulges and the hole goes oval. Below about two thicknesses the web can shear away entirely.

FIX Move the hole inboard to at least 2t from the edge, or pierce it before the blank is trimmed.

SOURCE Sheet-metal design guidance (minimum edge distance for a pierced hole, typically 2t).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 2.5 Steel DP600 (dual-phase) **>= 3** Steel DP980 (dual-phase) **>= 2.5** Steel (high-strength)

09 Holes too close together

MEDIUM

```
minHoleToHoleToThickness  >=  2 gap/t
sm-hole-to-hole
```

The web of material between two punched holes is pulled in both directions as each is pierced. Below about two material thicknesses it distorts, tears, or wears the punch and die out of tolerance early.

FIX Move the holes apart to at least 2t edge-to-edge, or pierce them in separate stations so the web is not loaded twice at once.

SOURCE Sheet-metal design guidance (minimum web between pierced holes, typically 2t).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 2.5 Steel DP980 (dual-phase) **>= 2.5** Steel (high-strength) **>= 2.5** Stainless Steel 304

10 Cut-out smaller than the material thickness

MEDIUM

```
minApertureToThickness  >=  1 size/t
sm-min-aperture
```

A punch narrower than the sheet is thick buckles under the piercing load, and the shape of the aperture does not change that — a 3 mm slot in 4 mm plate breaks the same punch a 3 mm hole would. This is the same limit `sm-hole-diameter` applies to round holes, extended to the slots, obrounds and shaped cut-outs that make up most of a real stamping.

FIX Open the smallest cut-out to at least one material thickness across, or laser it rather than piercing it and price the slower operation.

SOURCE Sheet-metal piercing guidance (punch slenderness), applied to the aperture's size rather than to a diameter it may not have.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 1.5 Steel DP980 (dual-phase) **>= 1.2** Stainless Steel 304

11 Springback beyond what overbend alone can compensate

MEDIUM

```
springbackOverbendPct  <=  8 % overbend required
sm-springback
```

Springback scales with YS/E, which is why high-strength steel recovers far more than mild steel at identical geometry. The book gives 2 to 8 percent added bend angle as the practical overbend allowance; past that the die has to bottom or stretch-bend the part, and the tooling is a different design at a different price.

FIX Tighten the bend radius, choose a lower-yield grade, or budget for bottoming or stretch bending rather than overbend alone.

SOURCE Boljanovic, Sec. 5.7, Eq. 5.22 for the springback factor and the 2-8% practical overbend allowance. READ FIRST-HAND.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

12 Punched hole smaller than the material thickness

MEDIUM

```
minHoleDiaToThickness  >=  1 d/t
sm-hole-diameter
```

A punch narrower than the sheet is thick is loaded in buckling and breaks often. It becomes a consumable rather than a tool.

FIX Increase the hole diameter to at least one thickness, or drill rather than punch.

SOURCE Sheet-metal design guidance (minimum punched hole diameter ~1x thickness).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · CONTESTED - SEE APPENDIX B

BY MATERIAL GRADE

>= 1.5	Steel DP980 (dual-phase)	**>= 2.5**	Steel 22MnB5 (press-hardened)	**>= 1.2**	Stainless Steel 316L
>= 1.2	Stainless Steel 304	**>= 1.5**	Steel (high-strength)	**>= 1**	Aluminium 6061

Deep drawing (multi-stage)

5 RULES

01 Too deep to draw in one operation

HIGH

```
drawDepthToWidth  <=  0.75 depth/width
dd-draw-depth
```

A cup deeper than about three quarters of its width cannot be drawn in one hit: the wall thins until it splits at the punch nose. Past that limit the part needs redraws, and every redraw is another die, another station and an anneal if the alloy work-hardens.

FIX Reduce the depth, widen the part, or plan for redraws — the first draw takes the most, and each redraw after it takes 1.3-1.5:1, not 2:1.

SOURCE Deep drawing design guidance: maximum depth-to-diameter in a SINGLE operation is 0.5-0.75 for most metals, approaching or slightly exceeding 1.0 for highly ductile low-carbon steel and aluminium; limiting draw ratio 1.8-2.3 for most steels, redraw ratios 1.3-1.5:1 per stage. MEASURED HERE AS A PROXY: this rule reads bounding-box depth over the narrower box width, not blank diameter over punch diameter — the blank and the punch are not in a finished solid. See UNWRITTEN_RULES for what that proxy cannot see. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

<= 1	Steel (mild)	**<= 0.6**	Aluminium 5052 (sheet)	**<= 0.5**	Steel (high-strength)

02 Cup too deep for the drawing-ratio table

HIGH

```
drawStagesBeyondTable  <=  0 beyond the table (1 = yes)
dd-draw-stages
```

Each draw can only reduce the diameter to m times the previous one, and Table 6.2 runs to five successive draws. A cup still larger than target after five is not simply expensive — it needs interstage annealing the table never contemplated, and the process plan is no longer a stamping plan.

FIX Reduce the draw depth, increase the starting blank thickness, or plan the part as a drawn-and-ironed or spun component.

SOURCE Boljanovic, Sec. 6.2.1 and Table 6.2: optimal drawing ratio m per operation against relative thickness 100*T/D. READ FIRST-HAND.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

03 Tolerance tighter than deep drawing holds

HIGH

```
tightestToleranceMm  >=  0.5 mm total band
dd-tolerance-capability
```

A drawn wall is formed between a punch and a die with clearance in between and it springs back when the punch withdraws. It is looser than a blanked edge, not tighter.

FIX Open the tolerance, or add an ironing or sizing station on the feature that needs it.

SOURCE Positioned above the 0.4 mm flat-stamping band used elsewhere in this catalogue, because a drawn wall adds springback the blanked edge does not have. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Undercuts cannot be drawn off the punch

HIGH

```
undercutFaceCount  <=  0 regions
dd-undercuts
```

The cup is stripped off the punch along the draw axis. A re-entrant feature holds it there, and the strip either tears the wall or leaves the part on the tool.

FIX Re-orient the part on the draw axis, or form the feature in a separate cam operation after the draw.

SOURCE Drawing tool design: the part is stripped along the draw axis, so a re-entrant feature cannot be produced by the draw itself. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Wall variation beyond what drawing produces

MEDIUM

```
wallSpreadRatio  <=  0.4 (p95-p5)/p50
dd-wall-uniformity
```

Drawing THINS the cup wall and thickens the flange, so a drawn part legitimately varies where a blanked one does not — but the variation comes from the draw, not from the drawing office. A section change the sheet never had is a machining feature on a pressing.

FIX Check whether the thick or thin region is a drawn consequence or a designed one; a designed section change on a drawn part is not a pressing.

SOURCE Deep drawing thins the wall progressively toward the punch nose while the flange thickens. The band here is positioned above the flat-sheet families in this catalogue and below the casting families; DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Fine blanking

6 RULES

01 Tolerance tighter than fine blanking holds

HIGH

```
tightestToleranceMm  >=  0.06 mm total band
fb-tolerance-capability
```

This is where fine blanking earns its tooling cost: a clean sheared face over the full thickness and a band an order of magnitude tighter than conventional stamping. It is not machining, and a band under about 0.06 mm total still belongs to a grinder.

FIX Open the tolerance, or grind that feature after blanking.

SOURCE Fine blanking is quoted as repeatably holding +/-0.01 to 0.03 mm depending on sheet type and thickness. The LOOSE end of that band (0.03 either way = 0.06 total) is used, so the rule flags only what is clearly outside the process. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Internal corner sharper than the sheet can be cut to

MEDIUM

```
minBendRadiusToThickness  >=  0.5 r/t
fb-corner-radius
```

A sharp internal corner concentrates the cutting stress into a point on the tool. The tool cracks there first, and the part carries a tear-out where a clean sheared face was specified.

FIX Put a radius of at least half the sheet thickness in every internal corner; an obtuse corner can go smaller, an acute one needs more.

SOURCE Fine blanking design guidance: minimum inside radius 0.5 x sheet thickness to spread stress evenly and avoid cracks; corner radius by angle — obtuse 5-10% of thickness, right angle 10-15%, acute 25-30%. The single 0.5t figure is used because the corner ANGLE is not measured by this engine (see UNWRITTEN_RULES). No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Pierced hole below the fine-blanking minimum

MEDIUM

```
minHoleDiaToThickness  >=  0.65 d/t
fb-hole-diameter
```

The V-ring grips the sheet and the counter-punch supports it, so the material does not flow away from the punch the way it does in conventional blanking. That is what lets a fine-blanked hole go below one thickness — but the punch is still a slender column under load and it breaks below about 0.65 of the sheet.

FIX Open the hole to at least 0.65 of the sheet thickness, or drill it as a secondary operation.

SOURCE Fine blanking design guidance: hole diameter approximately 60-65% of material thickness. NOTE — a second guide quotes minimum hole diameter >= 1.2 x thickness, which is nearly twice this figure and closer to the conventional-blanking rule. The two are in genuine conflict; the more permissive figure is used because going finer than conventional blanking is the capability fine blanking is bought for, and the finding should be treated as a screening result to put to your press shop. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Hole too close to the blanked edge

MEDIUM

```
minHoleToEdgeToThickness  >=  1.5 gap/t
fb-hole-to-edge
```

The V-ring clamps just outside the cut line, so the material between a hole and the edge is held far better than in conventional blanking — but it is still the narrowest section on the part and it is where the blank will bow if it goes too thin.

FIX Move the hole inboard to at least 1.5 thicknesses from the edge.

SOURCE Fine blanking design guidance: keep holes at least 1.5-2 x sheet thickness from the nearest edge, against 2t for conventional blanking. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Web between features narrower than the sheet is thick

MEDIUM

```
minHoleToHoleToThickness  >=  1 gap/t
fb-hole-to-hole
```

The web between two pierced features is unsupported on both sides while both punches are in the sheet. Below one thickness it collapses, and the two holes break through into one.

FIX Open the web to at least one sheet thickness, or combine the two features into one aperture.

SOURCE Fine blanking design guidance: maintain webs >= material thickness between adjacent features. Against 2t for conventional blanking. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Cut-out smaller than the fine-blanking minimum

MEDIUM

```
minApertureToThickness  >=  0.65 size/t
fb-min-aperture
```

A non-round aperture is cut by a shaped punch, and a slender punch snaps whatever its outline. The same floor applies as to a round hole.

FIX Open the aperture to at least 0.65 of the sheet thickness across its narrowest span.

SOURCE Fine blanking design guidance on minimum feature size, applied to non-circular apertures on the same basis as pierced holes. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Hot stamping / press hardening

5 RULES

01 Bend radius below what press-hardened steel tolerates

HIGH

```
minBendRadiusToThickness  >=  6 r/t
hs-bend-radius
```

The radius is formed hot and soft, but it is quenched to martensite in the same die and lives its life at 1500 MPa. A tight radius carries the residual stress of the quench on top of the forming strain, and it is where a hydrogen-assisted crack starts.

FIX Open the radius to at least six thicknesses, or move the bend to a region that stays soft (a tailored or partially-quenched blank).

SOURCE The 22MnB5 (press-hardened) figure carried by the sheet-metal bend rule in this same catalogue: 6 r/t. It is the base threshold here because hot stamping IS that alloy, so the two families cannot disagree about it. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Hole too small to pierce in press-hardened steel

HIGH

```
minHoleDiaToThickness  >=  2.5 d/t
hs-hole-diameter
```

A hole in a press-hardened part is either pierced hot before quench, when its position moves with the shrinkage, or cut by laser afterwards at a cost per hole. Neither route makes a small hole cheap, and a punch driven through martensite chips.

FIX Open the hole to at least 2.5 thicknesses, or accept laser trimming and cost it per hole.

SOURCE The 22MnB5 (press-hardened) figure carried by the sheet-metal hole rule in this same catalogue: 2.5 d/t. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than hot stamping holds

HIGH

```
tightestToleranceMm  >=  0.3 mm total band
hs-tolerance-capability
```

Quenching in the die is what makes hot stamping DIMENSIONALLY good — the part is held while it transforms, so there is almost no springback and very little warping. It is still a phase transformation inside a closing tool, and a band under about 0.3 mm total is a machining or laser requirement.

FIX Open the tolerance, or laser-cut that feature after quench.

SOURCE The 22MnB5 (press-hardened) figure carried by the sheet-metal tolerance rule in this same catalogue: 0.3 mm total band — TIGHTER than the 0.4 mm mild-steel stamping band, because quenching in the die removes the springback that drives cold-stamped variation. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Undercuts cannot be drawn from a quenching die

HIGH

```
undercutFaceCount  <=  0 regions
hs-undercuts
```

The die must stay closed on the part until the quench is complete and then open cleanly. An undercut needs a slide that has to be water-cooled like the rest of the tool and has to move against a part that is gripping it as it shrinks.

FIX Re-orient the part on the die axis, or cut the undercut feature by laser afterwards.

SOURCE Press-hardening tool design: the part is quenched in the closed die and then drawn, so the tool geometry must release a shrinking, hardening part. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Pierced holes too close for a hot-formed part

MEDIUM

```
minHoleToHoleToThickness  >=  1.5 gap/t
hs-hole-to-hole
```

Holes closer than about 1.5 thicknesses distort each other as they are pierced, and on a part that then shrinks through a quench the position error and the distortion add.

FIX Open the gap to at least 1.5 thicknesses, or loosen the position tolerance between them to cover springback and bend variation.

SOURCE Stamping design guidance: hole spacing of at least 1.5 x material thickness to avoid distortion; same-plane piercing holds close position, and holes closer than that or on different planes need a loosened band. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Roll forming

6 RULES

01 Tolerance tighter than roll forming holds

HIGH

```
tightestToleranceMm  >=  0.6 mm total band
roll-tolerance-capability
```

A roll-formed profile is worked gradually through many stations and springs back at every one, so cross-section tolerance is looser than a press brake and length tolerance depends on the flying cut.

FIX Open the profile tolerance and add a secondary operation for anything that must locate.

SOURCE Roll-forming dimensional tolerance guidance.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 0.8 Aluminium 6061

02 Hole too close to a formed corner

HIGH

```
holeToBendClearanceMm  >=  0 mm clear of 2t+r
roll-hole-to-bend
```

A hole pre-punched in the strip distorts as the corner is worked up over successive stations. Roll forming spreads the deformation over more stations than a press brake, which helps, but the hole still pulls oval if it sits inside the bend allowance.

FIX Move the hole clear of the forming zone, or pierce it after forming in a secondary station.

SOURCE Roll-forming design guidance (pre-punched features and the forming zone).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Section is not constant along the part

HIGH

```
wallSpreadRatio  <=  0.15 (p95-p5)/p50
roll-section-uniformity
```

A roll former makes ONE profile continuously and cuts it to length. It cannot vary the section along the run — that is the whole nature of the process, and it is why roll forming is cheap per metre. A part whose section changes cannot be roll formed at all.

FIX Hold one constant profile and put any variation into secondary punching, or move to press-brake forming or stamping.

SOURCE Roll-forming design guidance (constant cross-section is inherent to the process).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Inside bend radius tighter than the grade will take

MEDIUM

```
minBendRadiusToThickness  >=  1 r/t
roll-bend-radius
```

Roll forming works the bend up gradually over many stations, so it is gentler than a press brake — but the outer fibre still has to stretch, and below about 1 r/t it splits.

FIX Open the corner radius, or add stations so the bend is developed over a longer run.

SOURCE Roll-forming design guidance (minimum inside radius about 1 r/t for mild steel).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 3 Aluminium 6061 **>= 2** Steel (high-strength)

05 Flange too short for the rolls to grip

MEDIUM

```
minFlangeToThickness  >=  3 flange/t
roll-flange-length
```

A short flange has nothing for the rolls to hold and wanders, so the profile is out of tolerance along its length.

FIX Lengthen the flange to at least three material thicknesses, or form it in a secondary operation.

SOURCE Roll-forming design guidance (minimum flange length).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Pierced hole smaller than the material thickness

MEDIUM

```
minHoleDiaToThickness  >=  1 d/t
roll-hole-diameter
```

A punch narrower than the sheet is thick buckles under the piercing load. This is a punch-strength limit and it is the same in the strip as it is in a press.

FIX Open the hole to at least one material thickness, or drill it after forming.

SOURCE Sheet-metal piercing guidance (punch slenderness).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Hydroforming

5 RULES

01 Corner radius too tight for the forming pressure available

HIGH

```
minBendRadiusToThickness  >=  3 r/t
hydro-corner-radius
```

A hydroformed corner is filled by internal pressure pushing the wall into the die, and the pressure needed rises as the radius falls. Below about 3 r/t the tube thins and splits at the corner before the die is filled.

FIX Open the corner radii, or accept a two-stage form with an intermediate anneal.

SOURCE Tube hydroforming design guidance (corner radius against forming pressure and wall thinning).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 3 ferrous **>= 4** aluminium

02 Local section below the process minimum

HIGH

```
wallP5Mm  >=  0.8 mm
hydro-min-section
```

The tube thins where it expands, and the thinnest surviving section is what bursts. It is the measure that predicts the failure, not the average.

FIX Thicken the thinnest region, or move the feature to where the section is already full.

SOURCE Process minimum-section guidance. Checked on the 5th-percentile wall, which is the thinnest measured place rather than the nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Section varies more than the expansion limit allows

HIGH

```
wallSpreadRatio  <=  0.3 (p95-p5)/p50
hydro-section-uniformity
```

A hydroformed part starts as a tube of one wall thickness and thins wherever it expands. A large variation in the finished section means a large expansion ratio somewhere, and past roughly 30-40% expansion the wall splits.

FIX Reduce the largest expansion, start from a pre-bent tube closer to the final shape, or use a thicker starting gauge.

SOURCE Tube hydroforming design guidance (expansion ratio and wall thinning limits).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Tolerance tighter than hydroforming holds

HIGH

```
tightestToleranceMm  >=  0.5 mm total band
hydro-tolerance-capability
```

The tube conforms to the die under pressure, so the formed shape is repeatable, but the wall thins unevenly and the ends move as the tube is fed. About +/-0.25 mm is realistic.

FIX Tolerance from the die-formed faces, not from the tube ends.

SOURCE Tube hydroforming holds a permanent-die class of tolerance on the formed section because the tube is pressed against a steel die, but the tube diameter, wall and length going in all vary, and that variation carries through. The band sits between the stamping and casting families used elsewhere in this catalogue. DERIVED from that ordering; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Undercuts prevent the die from opening

HIGH

```
undercutFaceCount  <=  0 regions
hydro-undercuts
```

The formed tube is trapped in a two-part die that must open along one axis. An undercut locks the part in the tool.

FIX Re-orient the part relative to the die parting line, or split the feature so it forms between the halves.

SOURCE Hydroforming design guidance (die opening and part extraction).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Metal spinning

3 RULES

01 Part is not a body of revolution

HIGH

```
axisymmetricAreaPct  >=  90 % of surface area
spin-body-of-revolution  ·  BLOCKING
```

Spinning forms the part by pressing a roller against a blank rotating on a mandrel. Every surface it can make is a surface of revolution about the mandrel axis. A flat, a lug or a bolt boss is not reachable by that motion — not expensively, but at all.

FIX Spin an axisymmetric shell and add the non-axisymmetric features as a secondary operation, or press the part instead.

SOURCE Derived from the process kinematics rather than from a design guide: a rotating mandrel and a following roller generate only surfaces of revolution about the mandrel axis. The 90% threshold allows small secondary features on an otherwise round part; it is this tool's own screening value, shared with the centrifugal-casting rule for the same reason.

THIS TOOL'S OWN COST MODEL · NOT YET AUDITED

02 Tolerance tighter than spinning holds

HIGH

```
tightestToleranceMm  >=  0.8 mm total band
spin-tolerance-capability
```

The roller follows a path against a blank whose thickness is changing under it. CNC spinning is repeatable on round shapes, but the wall thins by a quarter to a third as it forms and the band reflects that.

FIX Open the tolerance, or machine the feature that needs it after spinning — the stock is there, because the blank was specified thicker.

SOURCE Metal spinning is quoted at tolerances of about 1/32 inch (0.79 mm), with material specified 25-30% thicker than the finished wall to allow for thinning. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Undercuts cannot be drawn off the mandrel

HIGH

```
undercutFaceCount  <=  0 regions
spin-undercuts
```

The finished shell is pulled off the mandrel along its axis. A re-entrant profile locks it on, and the only ways out are a collapsible mandrel — an expensive, wearing assembly — or scrapping the part.

FIX Remove the re-entrant profile, or budget for a collapsible or segmented mandrel.

SOURCE Metal spinning tool practice: the shell is drawn off the mandrel along the spin axis, so a re-entrant profile needs a collapsible mandrel. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

CHAPTER 4

Forging and bulk forming

Solid metal moved under load. Draft is far heavier than in casting, webs and ribs have hard minimums, and the die closes in one direction only.

27 RULES · 5 PROCESSES · 22 HIGH / 4 MEDIUM / 1 LOW · 6 CARRY MATERIAL-SPECIFIC THRESHOLDS

Forging (hot)

6 RULES

01 Wall area below the minimum hot-forging draft

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
forge-hot-draft-minimum  ·  cutoff 5 deg
```

A hot forging shrinks onto the die as it cools and is knocked out mechanically, so it needs far more draft than any casting — 3 degrees on external surfaces and 5-7 on internal ones is normal. This is the single largest difference between forging and casting geometry.

FIX Allow 3 degrees on external walls and 5-7 degrees on internal walls and pockets.

SOURCE Closed-die hot forging design guidance (draft typically 3 degrees external, 5-7 degrees internal).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 10 aluminium **<= 3** titanium

02 Web too thin to forge

HIGH

```
wallP5Mm  >=  3 mm
forge-hot-min-web
```

Metal must flow to fill a thin web, and a thin web chills against the die before it fills. It also drives forging load up sharply, because the thinner the web the higher the pressure needed to make the material move.

FIX Thicken the web to at least 3 mm, or machine the thin region after forging.

SOURCE Closed-die forging design guidance (minimum web thickness and forging load).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 2.5 aluminium **>= 3.5** ferrous **>= 5** titanium

03 Tolerance tighter than forging holds

HIGH

```
tightestToleranceMm  >=  1 mm total band
forge-hot-tolerance-capability
```

A hot forging shrinks as it cools, the dies wear, and the flash line moves with die closure. About +/-0.5 mm is the realistic as-forged band, and it grows across the parting line.

FIX Machine every functional surface. As-forged tolerances belong on the non-functional envelope only.

SOURCE Closed-die hot forging tolerance guidance (die wear and thermal contraction).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 0.6 aluminium **>= 1.4** titanium

04 Undercuts cannot be forged

HIGH

```
undercutFaceCount  <=  0 regions
forge-hot-undercuts
```

A forging die has two halves and no slides. There is no forging equivalent of a side action: an undercut simply cannot be forged and must be machined afterwards.

FIX Remove the undercut from the forged shape and machine it as a secondary operation, or re-orient the part on the parting line.

SOURCE Closed-die forging design guidance (two-part dies, no side actions).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Rib too tall for the metal to fill

MEDIUM

```
maxRibHeightToWall  <=  4 rib height / wall
forge-hot-rib-height
```

A tall, thin rib is the hardest thing to fill in a forging: the metal must flow furthest against the most die chilling, and an unfilled rib is scrap.

FIX Reduce the rib height, thicken it, or add generous fillets at its base so metal reaches the top.

SOURCE Closed-die forging design guidance (rib height-to-width and die filling).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Section varies more than the process tolerates

MEDIUM

```
wallSpreadRatio  <=  0.9 (p95-p5)/p50
forge-hot-uniformity
```

Metal flows from thick to thin under the die, and a large section change means the thin region fills last, against the most die chilling. It is also where the die cracks, because the pressure needed to fill it is highest.

FIX Even out the section and add generous fillets so metal reaches the thin regions.

SOURCE Casting and forging design guidance on gradual section change. A hot forging fills by displacing metal between closing dies rather than by flowing into a cavity, so it tolerates MORE section variation than a die casting and less than a machined part — this band sits between the two used elsewhere in this catalogue. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Forging (cold)

5 RULES

01 Web too thin for the forming load available

HIGH

```
wallP5Mm  >=  1.5 mm
forge-cold-min-web
```

Cold forging works the material below its recrystallisation temperature, so it work-hardens as it flows. A thin web needs very high pressure to fill and is where the tool cracks.

FIX Thicken the web, add an intermediate anneal, or machine the thin region afterwards.

SOURCE Cold forging design guidance (forming pressure and tool life against web thickness).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 2 ferrous	**>= 1.2** aluminium

02 Tolerance tighter than forging holds

HIGH

```
tightestToleranceMm  >=  0.2 mm total band
forge-cold-tolerance-capability
```

Cold forging holds far tighter than hot because nothing shrinks on cooling — about +/-0.1 mm — and that accuracy is most of the reason to pay for the tooling.

FIX Hold the tolerance as-forged; adding a machining operation gives away the advantage that justified the process.

SOURCE Cold forging / cold heading tolerance guidance.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Undercuts cannot be cold forged

HIGH

```
undercutFaceCount  <=  0 regions
forge-cold-undercuts
```

The part is ejected from a closed die by a knock-out pin along one axis. An undercut locks it in the tool.

FIX Machine the undercut as a secondary operation, or re-orient the part on the die axis.

SOURCE Cold forging design guidance (axial ejection).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Section varies more than the process tolerates

MEDIUM

```
wallSpreadRatio  <=  0.7 (p95-p5)/p50
forge-cold-uniformity
```

Cold forging work-hardens as it flows, so a large section change means the last region to fill is also the hardest to fill. Tool life is set by that region.

FIX Even out the section, or split the shape across two blows with an intermediate anneal.

SOURCE The same section-change guidance as hot forging, tightened because cold metal work-hardens as it moves: the load needed to fill a section change rises with every millimetre the metal travels, and the press has a limit the furnace removes in hot work. DERIVED from the hot-forging band in this catalogue, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Wall area below the minimum cold-forging draft

LOW

```
wallAreaBelowDraftPct  <=  10 % of wall area
forge-cold-draft-minimum  ·  cutoff 0.5 deg
```

A cold forging does not shrink onto the die the way a hot one does, and it is ejected by a knock-out pin rather than being drawn. Near-zero draft is normal, which is a large part of why cold forging holds tolerances that hot forging cannot.

FIX Allow 0.5 degrees where the function permits; zero-draft walls are usually acceptable here.

SOURCE Cold forging / cold heading design guidance (draft near zero because there is no thermal shrink onto the die).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Open-die forging

5 RULES

01 Wall area below the minimum open-die forging draft

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
odf-draft-minimum  ·  cutoff 7 deg
```

The workpiece is turned and repositioned between blows against tools that are flat or barely contoured. Nothing about that process produces a near-vertical wall, and asking for one means the wall is machined afterwards out of stock that has to be there.

FIX Open the taper to at least 7°, or accept that the face is a machined face and leave stock for it.

SOURCE Forging draft guidance, taken at the top of the hot-forging band (the closed-die family in this catalogue reads the 5° point). Open-die work is manipulated between simple tools and takes more taper than closed-die, so the 7° point is read instead. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Section thinner than open-die forging reaches

HIGH

```
wallP5Mm  >=  6 mm
odf-min-web
```

Between flat tools there is no cavity to fill and no die wall to hold the metal, so a thin web simply is not formed — it is machined out of a thicker one.

FIX Thicken the section, or move to closed-die forging where a cavity can hold a thinner web.

SOURCE Positioned at twice the closed-die hot-forging minimum used elsewhere in this catalogue (3 mm), because open-die work has no cavity to fill a thin section into. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than open-die forging holds

HIGH

```
tightestToleranceMm  >=  3 mm total band
odf-tolerance-capability
```

There is no die cavity setting the shape — the operator and the manipulator set it, blow by blow. Every dimension that matters on an open-die forging is machined afterwards, and the forging exists to put grain flow and mass roughly where they are needed.

FIX Treat the forging as a blank: put the tolerance on the machined feature and leave stock for it.

SOURCE Positioned at three times the closed-die hot-forging band used elsewhere in this catalogue (1.0 mm), because open-die work has no cavity setting the dimension. DERIVED from that ordering, not quoted — and it is the threshold in this family most in need of a forge-shop review.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Undercuts cannot be forged between open tools

HIGH

```
undercutFaceCount  <=  0 regions
odf-undercuts
```

Flat or simply-contoured tools approach along one axis and separate along it. There is no mechanism in an open-die press to form or release a re-entrant feature.

FIX Machine the undercut afterwards.

SOURCE Open-die forging tool practice: flat tools approach and separate along one axis. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Section change beyond what open-die work produces

MEDIUM

```
wallSpreadRatio  <=  1.2 (p95-p5)/p50
odf-uniformity
```

Open-die forging makes simple shapes: shafts, discs, rings, blocks. A part whose section varies more than the process can work into it is a machined part with a forged blank, and it should be quoted that way.

FIX Simplify the forged shape and machine the detail, or move to closed-die forging.

SOURCE Positioned above the closed-die hot-forging band used elsewhere in this catalogue (0.9), because open-die work is deliberately a simpler shape carrying more machining stock. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Cold heading / upsetting

5 RULES

01 Too large to cold head from wire

HIGH

```
maxEnvelopeMm  <=  150 mm largest dimension
ch-max-envelope  ·  BLOCKING
```

A cold header cuts a slug from coil and upsets it between dies. The machine is built around the wire it feeds, so the finished part cannot be much longer than the slug — commercial multi-station headers work in fasteners and small formed parts, not brackets.

FIX Cold heading is not a route for a part this size. Where a headed feature is wanted on a large part, head the fastener separately and join it.

SOURCE Cold-heading practice: production headers run wire up to roughly 25 mm diameter with cut-off lengths in the tens of millimetres; a 150 mm envelope is already generous for the largest machines. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 More metal than a header can cut as a slug

HIGH

```
partVolumeCm3  <=  80 cm3
ch-max-slug-volume  ·  BLOCKING
```

A cold header shears a slug off wire and upsets it. The slug is the whole part, so the part cannot hold more metal than the largest wire the machine runs can give it — about 25 mm diameter over a workable cut length.

FIX Cold heading is not a route at this mass. A forging or a casting is the comparable net-shape process.

SOURCE Cold-heading practice: production headers run wire to roughly 25 mm diameter; a 25 mm x 150 mm slug is about 73 cm3, so 80 cm3 is already the generous end. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than cold heading holds

HIGH

```
tightestToleranceMm  >=  0.2 mm total band
ch-tolerance-capability
```

Cold forming in a closed die is accurate and repeatable — that is why fasteners are made this way and not machined. Below about 0.2 mm total the feature is being ground or rolled, not headed.

FIX Open the tolerance, or grind the feature after heading.

SOURCE The cold-forging tolerance band used elsewhere in this catalogue (0.2 mm total), applied to cold heading as the same class of closed-die cold forming. DERIVED from that neighbour, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Undercuts cannot be ejected from a heading die

HIGH

```
undercutFaceCount  <=  0 regions
ch-undercuts
```

A headed part is knocked out of the die along the wire axis by a pin. A re-entrant feature holds it in the cavity, and heading machines run too fast to unload one part at a time.

FIX Remove the re-entrant feature, or machine it in afterwards — a cold-headed blank machines easily.

SOURCE Cold heading tool practice: the part is ejected along the die axis by a knock-out pin. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Head too tall to upset in one blow

HIGH

```
maxBossHeightToDia  <=  2.5 upset L/D
ch-upset-ratio
```

Cold heading forms the head by upsetting an unsupported length of wire. Past about two and a half wire diameters that free length buckles sideways instead of spreading, and the head folds over rather than filling.

FIX Reduce the head volume, or plan a two-blow sequence — a second blow reaches roughly four and a half diameters, and enclosing the blank in a die cavity extends the first.

SOURCE Cold heading practice: unsupported lengths of up to 2 to 2.5 x blank diameter can be upset in one blow in steel; enclosing the blank in a cavity of 1.5 x diameter allows more, and a two-blow sequence reaches about 4.5 diameters. MEASURED HERE AS A PROXY: the engine reads the recognised BOSS height over its diameter on the finished part, which is not the same thing as the free wire length before upsetting. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Extrusion

6 RULES

01 Wall thinner than the die can hold

HIGH

```
wallP5Mm  >=  1 mm
extr-min-wall
```

The die tongue that forms a thin wall is unsupported and deflects under extrusion pressure. Below about 1 mm in a solid aluminium profile the tongue chatters or breaks, and a hollow profile needs more again.

FIX Thicken the thinnest wall to 1.5-2 mm, especially on any hollow section.

SOURCE Aluminium extrusion design guidance: about 1.0 mm minimum for a solid profile, 1.5-2.0 mm for a hollow.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 1.5 copper

02 Section is not uniform enough for the die to run balanced

HIGH

```
wallSpreadRatio  <=  0.35 (p95-p5)/p50
extr-section-uniformity
```

Metal flows faster through a thick part of the die than a thin one. A profile whose wall varies sharply comes out twisted or bowed, and the die has to be corrected by trial — which is why extruders charge more for an unbalanced section and reject the worst of them outright.

FIX Even out the wall across the profile, and where a thick region is structurally necessary, blend into it rather than stepping.

SOURCE Aluminium extrusion design guidance (balanced metal flow and die correction).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 0.25 copper

03 Profile too large for a general-purpose press

HIGH

```
circumscribingCircleMm  <=  203 mm CCD
extr-circumscribing-circle
```

The circumscribing circle is the smallest circle the section fits inside, and it is what decides which press can run the job at all. Past about 203 mm (8 in) the profile leaves the general-purpose press population: it needs a bigger container, more tonnage and a specific supplier, and the quote changes shape rather than degrading gently.

FIX Bring the section inside a 203 mm circle, or split the profile into two extrusions and join them.

SOURCE Aluminium extrusion design guidance: extrusions are most economical within a circumscribing circle of roughly 1-10 in; general-purpose presses prefer CCD at or below 203 mm (8 in), with some plants reaching ~457 mm given tonnage and tooling. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Tolerance tighter than extrusion holds

HIGH

```
tightestToleranceMm  >=  0.5 mm total band
extr-tolerance-capability
```

An extruded profile is pulled through a die that wears and deflects, and it twists and bows as it cools on the run-out table. Straightness and twist usually bind before the section tolerance does.

FIX Tolerance the section loosely and specify straightness and twist separately; machine any locating feature.

SOURCE Aluminium extrusion tolerance guidance (EN 12020 / supplier standard practice).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Channel too deep for its width — the die tongue will fail

HIGH

```
maxPocketDepthToWidth  <=  3 depth/width
extr-tongue-ratio
```

A channel in the profile is a TONGUE in the die: a cantilevered finger of tool steel with metal flowing round it at hundreds of bar. It is the classic way an extrusion die breaks. Past about three times its own width the tongue deflects, the channel opens out along the run, and eventually the tongue snaps.

FIX Open the channel, shorten it, or put a large radius at its mouth and a full radius at its root — that buys roughly a 2:1 improvement.

SOURCE Aluminium extrusion design guidance: the ratio of channel height to width should be about 3:1 to keep die strength; with large radii at the opening and a full radius at the base a 2:1 improvement is available. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Feature blocks the profile leaving the die

HIGH

```
undercutFaceCount  <=  0 regions
extr-undercuts
```

An extrusion is a constant profile pushed through a fixed opening. Any feature that is not open along the extrusion axis cannot exist in the extruded shape and must be machined afterwards.

FIX Remove the feature from the profile and machine it after extrusion, or reconsider the extrusion axis.

SOURCE Extrusion design guidance (constant profile along the extrusion axis).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

CHAPTER 5

Machining and material removal

Material taken away by a tool. Cost follows access, setups and slenderness — how many directions the features are approached from, and how deep a tool has to reach.

24 RULES · 5 PROCESSES · 12 HIGH / 12 MEDIUM / 0 LOW · 4 CARRY MATERIAL-SPECIFIC THRESHOLDS

Machining (CNC mill/turn)

7 RULES

01 Thin web will chatter or deflect under the cutter

HIGH

```
wallP5Mm  >=  1.5 mm
```

mach-thin-web

A thin unsupported web deflects away from the tool, so the cut is inaccurate and the surface chatters. It usually forces a finishing pass at reduced depth of cut, or a support fixture.

FIX Thicken the web, add a rib, or leave sacrificial support material to be removed last.

SOURCE General machining design guidance (workpiece rigidity).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 0.8 ferrous	**>= 1** castiron	**>= 1.2** titanium
>= 1 aluminium	**>= 1.2** magnesium	**>= 2** plastic

02 Tolerance tighter than machining holds

HIGH

```
tightestToleranceMm  >=  0.05 mm total band
```

mach-tolerance-capability

A general machining tolerance of +/-0.025 mm (0.05 total) is routine on a rigid part. Tighter than that needs grinding, temperature control or in-process gauging, and each is a different cost base — not a tighter setting on the same machine.

FIX Open the tolerance to +/-0.025 mm or tighter only where the function needs it, and say WHICH feature needs it rather than applying a blanket tolerance to the drawing.

SOURCE General machining tolerance guidance (ISO 2768 fine class and shop practice).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

>= 0.08 titanium	**>= 0.2** plastic	**>= 0.05** aluminium

03 Deep pocket relative to its width

HIGH

```
maxPocketDepthToWidth  <=  4 depth/width
```

mach-pocket-depth-ratio

Beyond about 4:1 the cutter must be stepped down in a long, slender tool with reduced feed, and chip evacuation becomes the limiting factor. Cycle time rises faster than depth.

FIX Open the pocket out, split it across two setups, or accept a larger corner radius so a shorter, stiffer tool reaches the floor.

SOURCE General machining design guidance (tool L/D and chip-evacuation limits).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Surface a standard cutter cannot reach

HIGH

```
unreachableAreaPct  <=  2 % of surface area
mach-tool-access
```

A face a cutter cannot physically get to is not a machining problem to be priced — it is a feature that has to be made another way. The usual answers are EDM, a long-reach or custom tool, an extra setup, or a redesign, and all four cost more than the feature looks like it should.

FIX Open the pocket or slot so a standard cutter fits, increase the internal corner radii, or split the feature so it can be approached from a face that is already being machined.

SOURCE General machining design guidance (tool reach and shank clearance). Measured by sweeping a cylinder of the tool diameter along each approach direction — the HOLDER and machine envelope are not modelled, so this is a lower bound on the access problem, not an upper one.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Hole deeper than five diameters

MEDIUM

```
maxHoleDepthToDia  <=  5 depth/dia
mach-hole-depth-ratio
```

Past about 5x diameter a standard jobber drill stops clearing its own chips, so the cycle becomes a peck cycle — retract, clear, re-enter — and the tool wanders off centre. Past roughly 10x it is a gun-drilling or coolant-through operation on different equipment.

FIX Open the hole diameter, drill from both ends where the tolerance allows, or accept the deep-hole process and price it as one.

SOURCE General machining design guidance (chip evacuation and drill L/D limits).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL FAMILY

<= 6 aluminium	**<= 4** ferrous	**<= 5** castiron
<= 3 titanium	**<= 6** plastic	

06 Features approached from many directions

MEDIUM

```
setupCount  <=  3 setups
mach-setup-count  ·  PRICED BY THE COST ENGINE
```

Each additional setup adds fixturing, re-datuming and its own tolerance stack. Setup time is charged whether the batch is 10 parts or 1000.

FIX Group features onto fewer faces, or design a fixture that presents several faces in one clamping.

SOURCE Setup-driven cost is modelled directly in machining-feature-cost.mjs.

THIS TOOL'S OWN COST MODEL · NOT YET AUDITED

BY MATERIAL FAMILY

<= 2 titanium

07 Internal corner radius forces a small cutter

MEDIUM

```
minInternalCornerRadiusMm  >=  3 mm
mach-internal-corner-radius
```

An internal corner cannot be smaller than the cutter that makes it. A sharp internal corner on a drawing is either impossible or forces a very small, slow, fragile tool for the whole pocket.

FIX Specify the largest corner radius the function allows — often a third of the pocket depth — so a rigid cutter can be used.

SOURCE General machining design guidance (an internal corner radius equals the tool radius).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Turning (CNC lathe)

5 RULES

01 Shaft too slender to turn without extra support

HIGH

```
slendernessLtoD  <=  8 length/dia
turn-slenderness
```

A bar held in the chuck at one end deflects away from the tool and chatters. Past about three diameters it needs a tailstock, and past roughly eight it needs a steady rest — another setup, another fixture and a cycle that is no longer a simple turn.

FIX Shorten the part, increase the diameter, or accept the steady rest and cost the extra setup.

SOURCE Turning practice: a slender workpiece whips without support; a tailstock extends the workable ratio and a steady rest extends it further. The 8:1 figure is the commonly-quoted tailstock-supported limit. Reported from practitioner discussion rather than a design standard, and it is the threshold in this family most worth checking against your own machines.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Tolerance tighter than turning holds

HIGH

```
tightestToleranceMm  >=  0.02 mm total band
turn-tolerance-capability
```

A lathe holds a diameter better than a mill holds a position: the part spins on one axis and the tool moves on two. Below about 0.02 mm total the feature is ground or honed, not turned.

FIX Open the tolerance, or add a grinding operation and leave stock for it.

SOURCE Turning is quoted as holding tighter diametral tolerances than general milling. Positioned below the 0.05 mm general-machining band used elsewhere in this catalogue. DERIVED from that ordering; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Bore too deep for a boring bar

MEDIUM

```
maxHoleDepthToDia  <=  4 depth/dia
turn-hole-depth-ratio
```

A bore on a lathe is opened by a boring bar cantilevered from the turret. The bar has to fit inside the hole, so its own slenderness is set by the feature it is cutting, and past about four diameters it chatters and leaves a tapered bore.

FIX Open the diameter, shorten the bore, or use a damped or carbide bar and accept the tooling cost.

SOURCE Boring-bar practice: a steel bar is used to about 4:1 overhang and a carbide or damped bar beyond that. Positioned just below the general machining hole limit used elsewhere in this catalogue (5:1), because a boring bar is more constrained than a drill. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Internal corner sharper than a standard insert nose

MEDIUM

```
minInternalCornerRadiusMm  >=  0.4 mm
turn-internal-corner-radius
```

A turned internal corner is the radius of the insert nose that cut it. Standard ISO turning inserts come at 0.2, 0.4, 0.8 and 1.2 mm, and the smaller the nose the weaker the edge and the poorer the finish. Below 0.4 mm the corner is a grinding or EDM feature, not a turned one — and this is where turning beats milling by an order of magnitude, because a milled corner starts at the cutter radius.

FIX Open the corner to at least 0.4 mm, or accept a 0.2 mm nose and the reduced feed and edge life that come with it.

SOURCE Standard ISO turning insert nose radii (0.2 / 0.4 / 0.8 / 1.2 mm); the corner a turned feature carries is the nose radius of the insert that produced it. Widely published across insert catalogues; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Wall too thin to hold in a chuck

MEDIUM

```
wallP5Mm  >=  1.5 mm
turn-thin-web
```

A thin-walled turned part distorts in the chuck jaws before the tool touches it, and springs back out of round when it is released. The bore that measured perfectly in the machine is oval on the bench.

FIX Thicken the wall, or use soft jaws or an expanding mandrel and cost the fixture.

SOURCE The same thin-web limit the general machining family uses (1.5 mm), which is a cutting-force argument that applies equally to a lathe. Shared deliberately rather than restated with a different number.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Deep-hole / gun drilling

4 RULES

01 Hole deeper than a gun drill reaches

HIGH

```
maxHoleDepthToDia  <=  100 depth/dia
gundrill-hole-ld
```

A gun drill is a single-flute tool with coolant fed down its centre and chips carried out along an external vee. That is what lets it run to a hundred diameters where a twist drill is finished at five to ten. Past a hundred the tool wanders and the bore runs off, and the job needs BTA or a counter-bored approach from both ends.

FIX Shorten the hole, open the diameter, or drill from both ends and accept the step where they meet.

SOURCE Deep-hole drilling guidance: gun drills reach depth-to-diameter ratios of 100:1 and beyond, against roughly 5-10:1 for a conventional twist drill; past about 40:1 a dedicated deep-hole machine gives markedly better accuracy. Consistent across deep-hole machine builders; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Tolerance tighter than a gun-drilled bore holds

HIGH

```
tightestToleranceMm  >=  0.05 mm total band
gundrill-tolerance-capability
```

A gun-drilled bore is straight and round to a degree no twist drill approaches, but it is still a drilled hole a long way from its entry. A tighter band than the general machining limit means honing or reaming after it.

FIX Open the tolerance, or hone the bore and leave stock for it.

SOURCE The general machining band used elsewhere in this catalogue (0.05 mm total), applied unchanged because a gun-drilled bore is a drilled feature. Shared deliberately rather than restated with a different number.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Hole above the gun-drill diameter range

MEDIUM

```
maxHoleDiaMm  <=  30 mm dia
gundrill-max-dia
```

Above about 30 mm the chip volume outgrows the vee flute and the process changes: BTA takes the chips back through the middle of the tool instead. It is a different machine and a different quotation, not a bigger gun drill.

FIX Quote the large bore as BTA or trepanning; the small ones can stay on the gun drill.

SOURCE Gun drilling is quoted for diameters of about 1-30 mm; larger deep holes move to BTA or ejector drilling. Consistent across deep-hole machine builders; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Hole below the gun-drill diameter range

MEDIUM

```
minHoleDiaMm  >=  1 mm dia
gundrill-min-dia
```

A gun drill has to carry a coolant passage down its own centre, so there is a diameter below which there is no tool. Under about 1 mm the hole is EDM-drilled, not gun-drilled.

FIX Open the hole to at least 1 mm, or quote it as a fast-hole EDM feature.

SOURCE Gun drilling is quoted for diameters of about 1-30 mm, most commonly 2-20 mm. Consistent across deep-hole tooling suppliers; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Broaching

4 RULES

01 Hole too deep to broach

HIGH

```
maxHoleDepthToDia  <=  5 depth/dia
broach-hole-ld
```

The broach has to be longer than the feature by its whole cutting length, and it carries every chip it has cut along with it. Past about five diameters the tool is unwieldy, the chip packing is unmanageable and the pull load rises past what the machine will give.

FIX Shorten the feature, open it, or broach from both ends.

SOURCE Broaching process guidance: the depth-to-diameter ratio of a broached hole is generally not more than 5. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Blind features cannot be broached

HIGH

```
blindHoleCount  <=  0 blind features
broach-no-blind
```

A broach is pulled or pushed straight through the workpiece, each tooth taking a little more than the last. There is nowhere for it to stop and nowhere for the chips to go if it does. A blind feature is a milling, EDM or rotary-broaching job.

FIX Take the feature through, or use rotary broaching, which can form a blind profile to a shallow depth.

SOURCE Broaching process guidance: blind holes, deep holes, stepped holes and obstructed outer surfaces cannot be processed by broaching. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Feature above the broachable size range

MEDIUM

```
maxHoleDiaMm  <=  100 mm dia
broach-max-dia
```

Above about 100 mm the broach and the load needed to pull it both outgrow the standard machine, and the feature is bored or milled instead.

FIX Bore or mill the large feature; broaching stays worthwhile on the small ones.

SOURCE Broaching process guidance: broached hole diameters run generally from 10 to 100 mm. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Feature below the broachable size range

MEDIUM

```
minHoleDiaMm  >=  10 mm dia
broach-min-dia
```

A broach small enough for a fine hole has no cross-section left to carry the pull load, and it snaps inside the part.

FIX Open the feature, or produce it by wire EDM — which is what most small internal splines and keyways actually are.

SOURCE Broaching process guidance: broached hole diameters run generally from 10 to 100 mm. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Wire EDM

4 RULES

01 Blind features cannot be wire-cut

HIGH

```
blindHoleCount  <=  0 blind features
wedm-no-blind-features
```

Wire EDM threads a wire through a start hole and cuts a profile straight through the part. There is no way to stop part-way and leave a floor. A blind feature on a wire-cut part is a separate milling or sinker-EDM operation, and it is usually the one nobody quoted.

FIX Take the feature through, or quote the blind feature as a separate operation — sinker EDM if the corner needs to stay sharp, milling if it does not.

SOURCE Wire EDM process guidance: the wire cuts a through profile and cannot produce a blind pocket or hole. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Tolerance tighter than wire EDM holds

HIGH

```
tightestToleranceMm  >=  0.01 mm total band
wedm-tolerance-capability
```

Wire EDM is the most accurate route in this catalogue and this is where it earns its cycle time — but it is not unlimited, and below about 0.01 mm total the part is being jig-ground or lapped.

FIX Open the tolerance, or move that feature to grinding.

SOURCE Wire EDM is quoted as holding beyond +/-0.005 mm with multi-pass skim cutting. The total band used here is twice that single-sided figure. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Internal corner sharper than the wire can turn

MEDIUM

```
minInternalCornerRadiusMm  >=  0.15 mm
wedm-internal-corner-radius
```

The smallest inside corner a wire can leave is its own radius plus the spark gap — about 0.15 mm on a standard 0.25 mm wire. This is the number that decides whether a die plate is wire-cut or milled: an end mill cannot go below its own radius, and that is twenty times coarser.

FIX Open the corner to at least 0.15 mm, or specify a finer wire (0.10 mm reaches about 0.06 mm) and accept the slower cut.

SOURCE Wire EDM design guidance: minimum internal corner radius is approximately the wire RADIUS plus the spark gap — a 0.25 mm wire gives roughly 0.13-0.15 mm, and a 0.10 mm wire about 0.06 mm at reduced cutting speed. Consistent across several EDM suppliers; no primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Web too thin even for a no-force process

MEDIUM

```
wallP5Mm  >=  0.2 mm
```

wedm-thin-web

The wire exerts no cutting force, which is exactly why wire EDM leaves webs a milling cutter would tear off — the general machining limit of 1.5 mm does not apply here and applying it would reject a part the process handles easily. What does still apply is the flushing pressure and the part's own handling: below about 0.2 mm the web bends when the part is taken out of the tank.

FIX Thicken the web, or leave tabs and remove them after the part is out of the machine.

SOURCE Wire EDM removes material by spark erosion with no mechanical cutting force, so its thin-web limit is set by flushing and handling rather than by deflection. The value is positioned an order of magnitude below the 1.5 mm milling limit used elsewhere in this catalogue. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

CHAPTER 6

Polymers, rubber and composites

Melt or resin into a tool. Wall thickness governs both fill and cycle time, so it drives piece price twice over.

32 RULES · 5 PROCESSES · 15 HIGH / 13 MEDIUM / 4 LOW · 8 CARRY MATERIAL-SPECIFIC THRESHOLDS

Injection moulding

12 RULES

01 Wall thickness outside the practical moulding range

HIGH

```
wallP50Mm  in  1 to 3 mm
```

im-wall-thickness-range · PRICED BY THE COST ENGINE

Below about 1 mm the cavity is hard to fill before the melt freezes; above about 3 mm the part is cooling-limited, so cycle time climbs roughly with the square of the wall and sink marks appear over thick sections.

FIX Core out thick sections to a uniform nominal wall and add ribs for stiffness instead of solid mass.

SOURCE Injection-moulding design guidance; practical range for engineering thermoplastics.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

1.2-3.5 PC/ABS blend	**0.8-3** PBT	**1.2-4** PP-T20 (talc-filled)
1-3.5 PET	**0.8-3** PPS	**1-4** PEEK
1-4 TPU (thermoplastic PU)	**1-5** HDPE	**0.8-3** Polypropylene (PP)
1.2-3.5 ABS	**0.8-3** PA6 (Nylon)	**1.5-4** PA66-GF30 (glass-filled)
0.8-3 POM (Acetal)	**1-4** Polycarbonate (PC)	

02 Rib too thick at its base for the wall it stands on

MEDIUM

```
maxRibThicknessToWall  <=  0.6 rib t / wall t
```

im-rib-thickness-max · PRICED BY THE COST ENGINE

A rib meeting the wall at more than about 60% of the wall thickness makes a heavy junction that is the last place to solidify. It shows as a sink mark on the opposite — usually visible — surface, and as a void inside the section.

FIX Thin the rib to 40–60% of the nominal wall and add more ribs, or gusset it, if stiffness is lost.

SOURCE DuPont General Design Principles Module I, Fig 4.07 (p.26), READ FIRST-HAND from the page image: rib thickness T = 0.4 W for appearance parts, 0.6 W for structural parts (1.0 W only for foam molding), with 1/4-1/2 deg draft per side and a base fillet of half the rib thickness. The tool's 0.4-0.6 band matches the printed figures exactly - an audit CONFIRMATION, not a retune. The resin-specific tightenings above (0.5 for high-shrink semi-crystallines) are practising-moulder experience and are STRICTER than the printed 0.6, so they stand; Mack Molding's "60% is not sacred" caveat still applies on cosmetic faces.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

BY MATERIAL GRADE

<= 0.5 PBT	**<= 0.5** HDPE	**<= 0.6** PP-T20 (talc-filled)
<= 0.5 POM (Acetal)	**<= 0.5** PA6 (Nylon)	**<= 0.6** PA66-GF30 (glass-filled)
<= 0.6 ABS	**<= 0.5** Polycarbonate (PC)	**<= 0.5** PC/ABS blend
<= 0.5 TPU (thermoplastic PU)		

03 Wall area below the minimum draft angle

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
im-draft-minimum  ·  measured only on faces below 1 deg
```

A wall with no draft drags on the core as the part ejects, scuffing the surface and raising ejection force. Textured surfaces need considerably more draft than smooth ones.

FIX Add at least 0.5 to 1 degree per side on smooth walls; allow 2 to 5 degrees where the surface is textured. For DuPont resins the finding names the Table 3.01 angle it judged.

SOURCE Injection-moulding design guidance (0.5–1 deg/side smooth, 2–5 deg textured) for resins DuPont Table 3.01 does not name.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · PRIMARY SOURCE READ

BY MATERIAL GRADE

<= 5 PA6 (Nylon)	**<= 5** PA66-GF30 (glass-filled)	**<= 5** POM (Acetal)
<= 5 PET	**<= 5** Polycarbonate (PC)	**<= 5** PC/ABS blend
<= 5 TPU (thermoplastic PU)		

04 Tolerance tighter than injection moulding holds

HIGH

```
tightestToleranceMm  >=  0.2 mm total band
im-tolerance-capability
```

Moulding tolerance is dominated by shrinkage, which varies with fill, pack and the batch of resin. About 0.2 mm total on a 50 mm dimension is normal for an engineering thermoplastic; a filled or semi-crystalline grade is worse, not better.

FIX Open the tolerance, or machine the feature after moulding and price the secondary operation.

SOURCE Injection-moulding dimensional tolerance guidance (DIN 16901 / SPI commercial class).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

>= 0.22 PC/ABS blend	**>= 0.35** PBT	**>= 0.3** PP-T20 (talc-filled)
>= 0.5 HDPE	**>= 0.25** PEEK	**>= 0.25** PPS
>= 0.35 POM (Acetal)	**>= 0.35** PA6 (Nylon)	**>= 0.3** PA66-GF30 (glass-filled)
>= 0.2 ABS	**>= 0.2** Polycarbonate (PC)	

05 Inside corner radius below the DuPont stress-concentration knee

MEDIUM

```
resinFilletMargin  >=  1 x required fillet
im-internal-radius
```

Most plastics are notch sensitive, and the sharp internal corner is the leading cause of failure DuPont names first. Their stress-concentration curve flattens at R/T = 0.5 — half the wall is where extra radius stops buying anything — and even a "sharp" edge is allowed 0.5 mm.

FIX Open every internal corner to at least half the adjoining wall thickness, and never below 0.5 mm — the printed minimum that is "usually permissible even where a sharp edge is required".

SOURCE DuPont General Design Principles Module I, Fig 3.07 + p.7, READ FIRST-HAND: fillet radius = 1/2 wall thickness at the stress-concentration knee; minimum recommended corner radius 0.020 in (0.508 mm). DuPont-named resins only — the measure abstains on others.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

BY MATERIAL GRADE

>= 1 Polycarbonate (PC)	**>= 1** PC/ABS blend	**>= 1** TPU (thermoplastic PU)

06 Undercuts require side actions

HIGH

```
undercutFaceCount  <=  0 regions
im-undercuts
```

Any feature the two mould halves cannot form in a straight pull needs a slide, lifter or collapsible core. Each mechanism adds tooling cost, adds a moving part that can wear, and lengthens the cycle.

FIX Redesign the feature so it forms in the draw direction, relocate the parting line, or accept the side action with its tooling cost priced in. A SHALLOW beveled undercut may instead strip from the mould: DuPont Module I (pp.12-13) allows up to 5% for Delrin (circular shapes only), 6-10% for Zytel, and just 1-2% for glass-reinforced resins depending on mould temperature — the engine does not yet measure undercut depth %, so stripping feasibility is the engineer's call. Covestro (p.34) prints the same discipline for its families: stiff resins (Makrolon PC, Bayblend PC/ABS, Apec, filled) up to 2% with rounded 30-45 deg leading edges; Texin/Desmopan TPU 5% typically, 10% under ideal conditions.

SOURCE Injection-moulding design guidance (side actions add roughly $500–$5,000 of tooling per feature). Stripping limits per resin from DuPont Module I pp.12-13, read first-hand.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

07 Blind hole deeper than twice its diameter

MEDIUM

```
maxBlindHoleDepthToDia  <=  2 depth / dia
im-blind-core-depth
```

A blind hole is formed by a core pin supported at one end only — a cantilever the melt pushes sideways. Past twice the diameter the pin deflects and the hole comes out off-centre. Through holes escape the limit because the pin is supported at both ends.

FIX Keep blind holes to twice their diameter; deeper needs a stepped core pin or a counterbored wall to shorten the unsupported length (DuPont Fig 3.12), and the bottom should keep at least 1/6 of the diameter of material to avoid bulging.

SOURCE DuPont General Design Principles Module I, p.8 + Figs 3.12/3.16, READ FIRST-HAND: "the depth of a blind hole is generally limited to twice the diameter of the core pin."

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

08 Boss too tall for the core pin that forms it

MEDIUM

```
maxBossHeightToDia  <=  3 height/dia
im-boss-height
```

A tall boss is formed by a long, slender core pin standing alone in the cavity. The melt front pushes it sideways, so the hole drifts off position and the pin fatigues; the boss itself is also a thick section that sinks on the show surface behind it.

FIX Shorten the boss, or support it with gussets and blend it into a nearby wall so the core pin is not standing free.

SOURCE Injection-moulding design guidance (boss height <= 3x outside diameter; support tall bosses with ribs or gussets).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · PRIMARY SOURCE READ

BY MATERIAL GRADE

<= 2.5 PA66-GF30 (glass-filled) **<= 3** Polycarbonate (PC) **<= 5** PC/ABS blend
<= 5 TPU (thermoplastic PU)

09 Rib taller than the wall will fill and eject

MEDIUM

```
maxRibHeightToWall  <=  3 rib h / wall t
im-rib-height
```

A tall thin rib is a long dead-end for the melt and a deep, narrow slot in the tool steel that is hard to vent and hard to cool. It also grips the core on ejection, so it needs more draft than the rest of the part.

FIX Cap the rib at about 3x the nominal wall and use two or three shorter ribs, or a cross-rib, to recover the section modulus.

SOURCE Injection-moulding design guidance (rib height <= 3x nominal wall).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

BY MATERIAL GRADE

<= 2.5 POM (Acetal) **<= 2.5** PA6 (Nylon)

10 Rib too thin to fill reliably

LOW

```
minRibThicknessToWall  >=  0.4 rib t / wall t
im-rib-thickness-min
```

A rib much below 40% of the wall is a narrow, high-resistance flow path off the main cavity. It fills late or not at all, and a short-shot rib contributes none of the stiffness it was drawn for.

FIX Take the rib back up to 40% of the wall, or delete it and thicken the wall locally instead.

SOURCE DuPont General Design Principles Module I, Fig 4.07 (p.26), READ FIRST-HAND: the printed rib band starts at T = 0.4 W (design for appearance) - thinner is a high-resistance flow path that fills late or not at all. The 0.4 floor matches the printed figure exactly; resin-specific entries (PP 0.3, glass-filled nylon 0.5) are flow-behaviour experience and stand on their own grade.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

BY MATERIAL GRADE

>= 0.5	PA66-GF30 (glass-filled)	**>= 0.3**	Polypropylene (PP)

11 Non-uniform wall thickness

MEDIUM

```
wallSpreadRatio  <=  0.6 (p95-p5)/p50
im-wall-uniformity
```

Uneven walls cool at different rates, which drives differential shrinkage — the usual root cause of warp, sink and locked-in stress. Transitions should be tapered rather than stepped.

FIX Even out the nominal wall; where a change is unavoidable, blend it over about a 3:1 taper.

SOURCE Injection-moulding design guidance (3:1 transition rule).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

12 Boss outside diameter outside the 2–2.5× hole band

LOW

```
dupontBossOdToHole  in  2 to 2.5 boss OD / hole dia
im-boss-od
```

DuPont's band is two-sided: below twice the hole the boss lacks the wall to carry a screw or insert load; above 2.5x it is a heavy section that voids, sinks and stretches the cycle. The same band applies around molded-in inserts (Fig 3.27).

FIX Size the boss outside diameter between 2 and 2.5 times its hole; if more support is needed, gusset or rib the boss instead of thickening it.

SOURCE DuPont General Design Principles Module I, p.8 + Fig 3.27 (p.14), READ FIRST-HAND: "the outside diameter of a boss should be 2 to 2-1/2 times the hole diameter to ensure adequate strength." Judged on the worst coaxial boss/hole pair the recogniser finds; abstains when it finds none.

NAMED STANDARD, NOT READ FIRST-HAND · PRIMARY SOURCE READ

Rubber moulding

6 RULES

01 Local section below the process minimum

HIGH

```
wallP5Mm  >=  0.8 mm
rubber-min-section
```

Uncured rubber is stiff and slow-flowing. A local section below about 0.8 mm will not fill before the compound scorches.

FIX Thicken the thinnest region, or move the feature to where the section is already full.

SOURCE Process minimum-section guidance. Checked on the 5th-percentile wall, which is the thinnest measured place rather than the nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Tolerance tighter than rubber moulding holds

HIGH

```
tightestToleranceMm  >=  0.5 mm total band
rubber-tolerance-capability
```

Rubber shrinks heavily on cure and keeps moving afterwards, and it deforms under its own measurement. Tolerances are graded (RMA A/B/C) and even the closest is loose against a rigid part.

FIX Use a graded rubber tolerance class rather than a machined-part tolerance, and dimension in the free state.

SOURCE RMA rubber dimensional tolerance classes.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Wall thickness outside the practical rubber-moulding range

MEDIUM

```
wallP50Mm  in  1 to 6 mm
rubber-wall-thickness-range
```

Rubber cures from the outside in, and cure time rises with the square of the section — a thick part is a slow part and risks an under-cured core. Below about 1 mm the uncured compound will not fill reliably.

FIX Hold 2-4 mm as the nominal section and hollow out heavy masses.

SOURCE Rubber moulding design guidance (cure time against section thickness).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Non-uniform section cures unevenly

MEDIUM

```
wallSpreadRatio  <=  0.8 (p95-p5)/p50
rubber-wall-uniformity
```

The cure is set by the thickest section, so a part with one heavy region holds the whole press open while the thin regions over-cure and lose elongation.

FIX Even out the section, or hollow the heavy region.

SOURCE Rubber moulding design guidance (cure uniformity).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Wall area below the minimum rubber-moulding draft

LOW

```
wallAreaBelowDraftPct  <=  15 % of wall area
rubber-draft-minimum  ·  cutoff 0.5 deg
```

Rubber is elastic and strips off modest undercuts and zero-draft walls that would lock a rigid part in the tool. Draft helps but is rarely the constraint.

FIX Allow 0.5-1 degree where it is free; a zero-draft rubber wall is usually acceptable.

SOURCE Rubber moulding design guidance (elastic recovery permits low draft).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Undercuts complicate the tool

LOW

```
undercutFaceCount  <=  0 regions
rubber-undercuts
```

Rubber strips off undercuts that would lock a rigid part in the tool — that elasticity is a genuine design freedom. Each one still slows demoulding and eventually tears the part, so the count is a cycle-time driver.

FIX Keep undercuts shallow enough to strip, or split the tool where they are deep.

SOURCE Process tooling guidance (split lines, loose pieces and cores).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Thermoforming (vacuum / pressure)

4 RULES

01 Wall area below the minimum thermoforming draft

HIGH

```
wallAreaBelowDraftPct  <=  5 % of wall area
tf-draft-minimum  ·  cutoff 2 deg
```

The sheet shrinks onto the tool as it cools. On a male tool it grips hard and needs 4-6 degrees; on a female tool it pulls away and 1.5-2 will do. Two degrees is the floor either way, and a vertical wall on a male tool will not come off at all.

FIX Add at least 2° to every drawn wall — and 4-6° if the part forms over a male tool rather than into a cavity.

SOURCE Thermoforming design guidance: 1.5-2° on vertical female features (part formed into the mould) and 4-6° on male features (part formed over the mould). The lower figure is used as the floor because this engine cannot tell male tooling from female. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Draw deeper than the sheet will stretch evenly

HIGH

```
drawDepthToWidth  <=  1 depth/width
tf-draw-ratio
```

The sheet is one thickness before forming and it has to cover the whole tool. Everything drawn deep is drawn thin, and past a depth equal to the width the corners of the part carry a fraction of the wall the flange still has. Beyond this the part needs plug assist — a second tool that pre-stretches the sheet — and that is a different quotation.

FIX Reduce the depth, widen the part, or accept plug-assisted forming and cost the plug.

SOURCE Thermoforming design guidance: for depth-to-width ratios exceeding 1:1, plug-assisted female forming is normally needed to keep material distribution acceptable; 3:1 is quoted as the outer limit. The 1:1 point is used because it is where the PROCESS changes rather than where it fails. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than thermoforming holds

HIGH

```
tightestToleranceMm  >=  1 mm total band
tf-tolerance-capability
```

Only one face of a thermoformed part touches the tool; the other is free and its position depends on how the sheet happened to thin. Anything tight is trimmed or machined afterwards, not formed.

FIX Put the tight dimension on the tool side, or trim it after forming.

SOURCE Thermoforming is the loosest of the plastic-forming routes in this catalogue: the sheet contacts one tool face only and the wall thins unevenly. Positioned above the rotational-moulding and injection-moulding bands used elsewhere here. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Undercuts trap the sheet on the tool

HIGH

```
undercutFaceCount  <=  0 regions
tf-undercuts
```

A formed sheet is stripped off the tool along the draw axis while it is still warm. A re-entrant feature holds it, and the part tears or stays on the tool.

FIX Remove the re-entrant feature, or use a split or collapsible tool and cost it.

SOURCE Thermoforming tool practice: the part is stripped along the draw axis. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Rotational moulding

4 RULES

01 Wall thinner than rotational moulding can build

HIGH

```
wallP5Mm  >=  3 mm
rm-min-wall
```

The wall is built by powder sticking to a hot mould, tumble after tumble. There is no pressure to push it anywhere. Below about 3 mm the coverage is a matter of luck, and the risk is not a thin wall but a hole.

FIX Thicken the wall to at least 3 mm, or blow-mould the part instead.

SOURCE Rotational moulding design guidance: use a minimum wall of 3 mm and a maximum of 10 mm; below roughly 1.9 mm (0.075 in) areas of the part risk inadequate or zero thickness. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Undercuts cannot be pulled from a rotational mould

HIGH

```
undercutFaceCount  <=  0 regions
rm-undercuts
```

A rotational mould is a shell that splits and is opened by hand. There is no ejection mechanism and no slide, so a re-entrant feature has to come off by flexing the part — which works on a thin flexible wall and not on anything else.

FIX Remove the re-entrant feature, or design it shallow enough that the part flexes off.

SOURCE Rotational moulding tool practice: split shell moulds opened manually, with no ejection system. Derived from the process, not quoted from a design guide.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Wall area below the minimum rotational-moulding draft

MEDIUM

```
wallAreaBelowDraftPct  <=  5 % of wall area
rm-draft-minimum  ·  cutoff 1 deg
```

The part shrinks onto the mould as it cools and has to be pulled off by hand. A degree or two of taper is what makes that possible without a puller.

FIX Add 1-2° of draft to every vertical wall; more on a textured surface.

SOURCE Rotational moulding design guidance: draft angles of generally 1-2°, varying with the shrinkage of the polymer. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Wall thicker than the cycle will fuse

MEDIUM

```
wallP50Mm  <=  10 mm
rm-max-wall
```

Heat reaches the powder through the mould, so a thick wall means a long cycle and an inner surface that is still fusing while the outer one degrades. This is one of the few processes with a MAXIMUM wall as well as a minimum.

FIX Thin the wall and add ribs or a kiss-off, or move to a pressure process.

SOURCE Rotational moulding design guidance: minimum 3 mm, maximum 10 mm nominal wall. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Composite layup / RTM

6 RULES

01 Local section below the process minimum

HIGH

```
wallP5Mm  >=  1 mm
rtm-min-section
```

A laminate cannot be thinner than the plies it is built from. Below about 1 mm there are too few plies to lay up repeatably and the resin flow front stalls.

FIX Thicken the thinnest region, or move the feature to where the section is already full.

SOURCE Process minimum-section guidance. Checked on the 5th-percentile wall, which is the thinnest measured place rather than the nominal.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Tolerance tighter than composite layup / rtm holds

HIGH

```
tightestToleranceMm  >=  0.5 mm total band
rtm-tolerance-capability
```

A laminate takes the tool surface accurately on the tool side and the bag or second tool side less so, and it springs in as it cures off a corner. Thickness varies with fibre volume fraction.

FIX Tolerance from the tool face, and allow for spring-in at every corner.

SOURCE Composite moulding tolerance guidance (tool-side vs bag-side, cure spring-in).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Undercuts trap the cured part in the tool

HIGH

```
undercutFaceCount  <=  0 regions
rtm-undercuts
```

A cured laminate has no give. An undercut needs a split or collapsible tool, which multiplies tooling cost and cycle time.

FIX Re-orient the part on the tool split line, or bond the undercut feature on as a secondary part.

SOURCE Composite tooling design guidance (split tools and collapsible cores).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Wall area below the minimum RTM draft

MEDIUM

```
wallAreaBelowDraftPct  <=  5 % of wall area
rtm-draft-minimum  ·  cutoff 1 deg
```

A cured laminate is stiff and grips a matched tool hard. Without draft it cannot be released without damaging the part or the tool surface.

FIX Allow 1-3 degrees on all tool-contact surfaces.

SOURCE Composite tooling design guidance (part release from matched moulds).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Section varies more than the process tolerates

MEDIUM

```
wallSpreadRatio  <=  0.8 (p95-p5)/p50
rtm-uniformity
```

Resin flows fastest through the thickest part of the preform, so an uneven laminate traps dry spots at the flow-front convergence. Thickness change is also where a laminate delaminates.

FIX Even out the ply count, and taper thickness changes over at least 20:1 rather than stepping them.

SOURCE Composite design guidance on uniform laminate thickness. A resin-transfer laminate is built from plies of a fixed thickness, so a section change is a ply drop — a stress raiser and a resin-rich pocket. The band sits with the bulk-forming families in this catalogue rather than the casting ones. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Laminate thickness outside the practical RTM range

MEDIUM

```
wallP50Mm  in  1.5 to 10 mm
rtm-wall-thickness-range
```

A laminate is built from plies of finite thickness, so a very thin wall has too few plies to be laid up repeatably, and a very thick one is slow to wet out and exotherms as it cures.

FIX Hold 2-5 mm and vary stiffness through the lay-up and ply orientation rather than through thickness.

SOURCE RTM / resin-transfer moulding design guidance (ply count and cure exotherm).

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

CHAPTER 7

Powder and additive

Built up rather than cut down. Density, aspect ratio and — for additive — build orientation replace the usual mould-opening constraints.

17 RULES · 3 PROCESSES · 13 HIGH / 4 MEDIUM / 0 LOW · 0 CARRY MATERIAL-SPECIFIC THRESHOLDS

Powder metallurgy (press & sinter)

6 RULES

01 Wall thinner than the tooling will survive

HIGH

```
wallP5Mm  >=  1.52 mm
pm-min-wall
```

A thin wall in the part is a thin blade in the die, and that blade takes the full compaction pressure on every stroke. The limit is set by tool life, not by the powder.

FIX Thicken the wall to at least 1.5 mm; a long thin wall is worse than a short one at the same thickness.

SOURCE Powder metallurgy design guidance: wall thickness should not be less than 1.52 mm (0.060 in), and long thin walls make the tooling fragile and shorten its service life. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Powder column too deep for its wall — density will vary

HIGH

```
pressDepthToWallRatio  <=  8 depth/wall
pm-press-depth-ratio
```

Powder does not flow. It is compacted along one axis and friction against the die wall means the density falls away from the punch face, so a deep column in a thin wall comes out dense at the top and soft at the bottom. That gradient is not visible, it does not sinter out, and it is measured as a strength difference in the finished part.

FIX Reduce the depth, thicken the wall, or split the part — a double-action press buys some depth but does not remove the gradient.

SOURCE Powder metallurgy design guidance: density variations become unavoidable once the length-to-wall-thickness ratio passes about 8:1. Related figures in circulation are an overall aspect ratio limit of 3:1 even with double-ended compaction, and L/D limits of 2 for a single-action press and 4 for a double-action one. The 8:1 wall figure is used because it is the one stated against WALL thickness, which is what this measure reads. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Features approached from more than the press axis

HIGH

```
setupCount  <=  1 directions
pm-single-press-axis
```

Compaction and ejection both happen along one vertical axis. A feature that needs a second approach direction — a cross hole, a side pocket, a transverse slot — cannot be pressed at all. It is a machining operation on a sintered blank, and it is usually the one missing from the quotation.

FIX Move the feature onto the press axis, or quote the cross-feature as secondary machining.

SOURCE Powder metallurgy design guidance: because powder compaction and ejection both occur along a vertical axis, there can be no obstructions in the press direction; undercuts and transverse holes cannot be formed directly and must be produced by secondary machining. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Tolerance tighter than pressing and sintering holds

HIGH

```
tightestToleranceMm  >=  0.1 mm total band
pm-tolerance-capability
```

In the press plane the die sets the dimension and PM is very repeatable — that is why it competes with machining at all. Along the press axis the fill and the sintering shrinkage both move it. A band under about 0.1 mm total is sized or machined afterwards.

FIX Open the tolerance, or add a sizing operation and cost it.

SOURCE Powder metallurgy is dimensionally repeatable in the press plane and less so along the press axis, where fill variation and sintering shrinkage act. Positioned between the cold-forging and general-machining bands used elsewhere in this catalogue. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Undercuts cannot be pressed

HIGH

```
undercutFaceCount  <=  0 regions
pm-undercuts
```

The same single-axis argument, in the other direction: the compact is ejected by the punch that formed it and nothing can be re-entrant to that motion.

FIX Remove the undercut, or machine it into the sintered part.

SOURCE Powder metallurgy design guidance: no obstructions in the press direction; undercuts must be produced by secondary machining. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Hole too small to press

MEDIUM

```
minHoleDiaMm  >=  1.5 mm dia
pm-min-hole
```

A hole in a pressed part is a core rod standing in the die through the whole compaction stroke. Below about 1.5 mm the rod is too slender to survive the pressure cycle and it is drilled into the sintered blank instead — a secondary operation on a process bought for being net-shape.

FIX Open the hole to at least 1.5 mm, or quote the drilling as a secondary operation.

SOURCE Powder metallurgy design guidance: the minimum recommended hole diameter is 1.5 mm (0.060 in), alongside a minimum wall of 1.52 mm. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Metal injection moulding (MIM)

7 RULES

01 Too large for metal injection moulding

HIGH

```
maxEnvelopeMm  <=  100 mm largest dimension
mim-max-envelope  ·  BLOCKING
```

A MIM part is moulded from powder in a binder and then sintered, shrinking by around 15-20%. That shrink has to be held to a tolerance across the whole part, so size is limited by distortion and by what a debind furnace can support — the process lives in small, complex parts.

FIX Above roughly 100 mm the route is a casting or a machined blank. Where MIM is wanted for one intricate feature, make that feature a separate MIM insert.

SOURCE MIM practice: typical parts are under 100 g and under 100 mm, limited by sintering shrinkage control and debind support. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Too much metal to debind and sinter

HIGH

```
partVolumeCm3  <=  30 cm3
```

mim-max-volume · BLOCKING

A MIM part loses its binder and then densifies by 15-20% in the furnace. The larger the mass, the further any point of it has to move, and the harder that shrink is to hold — which is why the process lives in small, intricate parts rather than large ones.

FIX Above roughly 30 cm3 the route is a casting or a machined blank; a MIM insert can still carry one intricate feature of a larger assembly.

SOURCE MIM practice: typical production parts are well under 100 g (about 13 cm3 in steel), with the upper end around 250 g. 30 cm3 is the generous end of that band. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Section too thick to debind

HIGH

```
wallP50Mm  <=  12.5 mm
```

mim-max-wall

Every gram of binder in the middle of the part has to travel out through the part to get away. Past about 12.5 mm the centre is still debinding when the outside has started to sinter, and the result is a part with a core defect nobody can see.

FIX Core out the heavy section, or split the part — 1.3 to 6.3 mm is where MIM is comfortable.

SOURCE MIM design guidance: wall thicknesses of 1.3-6.3 mm are preferred, and sections thicker than 12.5 mm should be avoided. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Wall thinner than a MIM feedstock will fill

HIGH

```
wallP5Mm  >=  0.5 mm
```

mim-min-wall

A MIM feedstock is a metal powder in a binder — far more viscous than a neat polymer. Below about 0.5 mm it will not fill, and what does fill will not survive debinding.

FIX Thicken the section to at least 0.5 mm.

SOURCE MIM design guidance: minimum wall thickness 0.5 mm. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

05 Tolerance tighter than MIM holds

HIGH

```
tightestToleranceMm  >=  0.1 mm total band
```

mim-tolerance-capability

A MIM part shrinks by around a fifth on sintering, uniformly if everything went right. The tolerance is a percentage of the dimension, so a band that is generous on a 40 mm feature is impossible on a 4 mm one.

FIX Open the tolerance, or machine the one feature that needs it after sintering.

SOURCE MIM tolerances are quoted as a PERCENTAGE of dimension (of order +/-0.3%), which this engine cannot express — it reads one tightest band in millimetres. The flat value here is a screening figure only; see UNWRITTEN_RULES for what a percentage-of-dimension rule would need. DERIVED, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

06 Internal corner sharper than a MIM part will survive

MEDIUM

```
minInternalCornerRadiusMm  >=  0.2 mm
```

mim-corner-radius

A sharp internal corner is a stress raiser in the green part, which is chalk-fragile between moulding and sintering. It cracks there while it is being handled, long before anyone loads it.

FIX Put a radius of at least 0.2 mm in every internal corner.

SOURCE MIM design guidance: corner radius greater than 0.2 mm. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

07 Undercuts need a slide or a collapsible core

MEDIUM

```
undercutFaceCount  <=  0 regions
mim-undercuts
```

MIM CAN mould an undercut — with a slide, or a collapsible core — which is the one thing it does that press-and-sinter cannot. What it cannot do is make it cheap: the mechanism costs tooling and brings flash with it.

FIX Remove the undercut, or accept the slide and cost it — it is still likely cheaper than the machining press-and-sinter would need.

SOURCE MIM design guidance: undercuts can be moulded using slides or collapsible cores — an advantage over conventional powder metallurgy — but the added cost and potential flashing mean internal undercuts are avoided in most MIM designs. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

Laser powder bed fusion (DMLS / SLM)

4 RULES

01 Wall thinner than the melt pool can stand up

HIGH

```
wallP5Mm  >=  0.4 mm
lpbf-min-wall
```

A wall a laser-track wide has nothing to conduct heat into but the track below it. It warps as it builds and it may not survive depowdering.

FIX Thicken the wall to at least 0.4 mm, and to 0.6 mm if it is tall or free-standing.

SOURCE Powder-bed fusion design guidance: a strut or thin-wall diameter of at least 0.4-0.6 mm should be chosen. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

02 Downward surface too shallow to build without support

HIGH

```
overhangAreaBelowDeg  <=  10 % of surface area
lpbf-overhang-45
```

A downward-facing surface is built onto loose powder, and loose powder does not conduct heat away. Below about 45 degrees from the plate the melt pool sinks in, the surface dross-forms and the geometry drifts — so the region needs a support, which has to be built, cut off, and the witness dressed. Support is not a small cost: it is often the largest single line in an AM quotation.

FIX Re-orient the part on the plate, add a self-supporting chamfer to shallow faces, or re-model horizontal holes as teardrops — a teardrop whose top angle stays at or above 45° builds at almost any diameter.

SOURCE Powder-bed fusion design guidance: 45° is the widely-used self-supporting threshold, and teardrop features with a 45° top angle show no significant material fall-in. Material and parameter dependent — some alloy and parameter sets self-support below this, and lattice struts want better than 25°. The rule reads the 45° point off a curve so a different angle can be asked for. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

03 Tolerance tighter than an as-built surface holds

HIGH

```
tightestToleranceMm  >=  0.2 mm total band
lpbf-tolerance-capability
```

An as-built powder-bed surface carries partly-melted powder and the part moves as residual stress relaxes on cutting from the plate. Every tight feature on an AM part is machined afterwards, and the stock for it has to be in the model.

FIX Open the tolerance, or add machining stock on that feature and quote the operation.

SOURCE As-built powder-bed fusion surfaces are quoted well looser than machined ones, with diameter deviation significant below Ø4 mm. Positioned above the general machining band used elsewhere in this catalogue. DERIVED from that ordering, not quoted.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

04 Hole too small to build open

MEDIUM

```
minHoleDiaMm  >=  1.5 mm dia
lpbf-min-hole
```

A hole below about 1.5 mm closes as the melt pool bridges it, and whatever does stay open is full of partly-sintered powder that has to be got out. Under 4 mm the diameter deviates enough that a fit is not safe to specify as-built.

FIX Open the hole to at least 1.5 mm — and to 4 mm if the diameter matters — or build it undersize and ream it.

SOURCE Powder-bed fusion study data: holes at Ø1.5 mm remained fully open, while holes below Ø4 mm showed high diameter deviation. Alloy dependent. No primary standard audited.

INDUSTRY CONSENSUS, NO PRIMARY SOURCE AUDITED · NOT YET AUDITED

APPENDIX A

What this book does not check

Every entry above is something the engine measures. This appendix is the other half of the truth: limits that matter to a manufacturing engineer and that this tool cannot currently evaluate honestly. Each one names what would be needed to close it and what the tool does instead in the meantime. It is here because a rule book that lists only what it can do teaches the reader to assume the rest was covered.

01 Cast-in-place insert wall thickness (magnesium)

Recognition of an INSERT — a body of a different material cast into the part. NADCA §3.2 p.63 gives a checkable limit: "Insert wall thicknesses 0.050 in. (1.25 mm) or less will heat and subsequently contract, and limit residual stresses to safe levels. Inserts with thicker walls must be preheated." A thick-walled insert in AZ91 is a stress-corrosion-cracking risk. A STEP file of a single solid carries no insert, so there is nothing to measure.

MEANWHILE Nothing. This rule cannot be evaluated from a one-body model at all, and no rule in the catalogue approximates it.

02 INSIDE versus OUTSIDE surfaces for the NADCA draft split

A test that tells an ejector-half surface from a cover-half one and survives through-features. NADCA requires TWICE as much draft on inside surfaces as on outside ones, because the alloy shrinks onto the features that form inside surfaces and away from those that form outside surfaces. The obvious ray test — does the outward normal escape to open air — classifies a through-hole wall as an outside surface, because from a bore wall the normal escapes through the bore opening.

MEANWHILE Every wall is judged at the OUTSIDE constant, which is the lower of the two. The draft rules therefore UNDER-report on ejector-half surfaces by a factor of two, and each of their source texts says so.

03 Parting-line planarity, and the trim die it pays for

The silhouette of the part along the chosen draw direction, and a measure of how far it departs from a single plane. NADCA §2.3 p.30: "Where possible, the die casting should be designed with the parting line in one plane to simplify trim die design", and trim dies "can be estimated at an additional 15 to 20% of the die casting die cost". A stepped parting line is a real and quotable cost.

MEANWHILE The draw direction IS chosen by sweeping candidate axes and scoring undercut area, so the parting direction is measured — but nothing reports whether the resulting parting line is planar, and no trim-die line appears in the tooling cost.

04 Tolerance expressed as a PERCENTAGE of dimension (MIM, and casting CT grades)

A per-feature tolerance chain rather than one tightest band. MIM is quoted at roughly +/-0.3% of the dimension and ISO 8062 casting grades are the same idea — a band that is generous on a 40 mm feature is impossible on a 4 mm one. This engine reads ONE tightest callout in millimetres from the PMI, so it cannot ask "is this band achievable FOR THIS SIZE".

MEANWHILE `mim-tolerance-capability` and every `*-tolerance-capability` rule use a flat millimetre screening value and say so. A part whose tight band sits on a small feature will pass when it should fail.

05 Best BUILD ORIENTATION for an additive part

A sweep over candidate build directions, scoring support area, height and down-facing area for each — the additive equivalent of the draw-direction sweep this engine already does for moulding. `overhang()` measures the part AS DRAWN along +Z, and re-orienting on the plate is the first thing an AM engineer does.

MEANWHILE The overhang figure is a measure of the model in its modelled orientation, and the rule says so. A part that fails at 45° may be entirely self-supporting once tipped.

06 BLOW MOULDING as a family

Sourced numbers for the blow ratio and for pinch-off geometry. The research turned up neither, and blow moulding competes directly with rotational moulding — a family built from rotomoulding's numbers with a different name would make the comparison between them meaningless, which is exactly what the route table exists to support.

MEANWHILE None. Blow moulding is not offered; rotational moulding is, and carries the hollow-part question for now.

07 SINKER (ram) EDM as its own family

Sourced numbers for the two limits that would distinguish it: the smallest internal corner a shaped electrode can burn, and the deepest rib an electrode can carry without arcing. The research found neither. Sinker EDM is the exact inverse of wire EDM — it is the process that CAN make a blind pocket with a sharp corner — so a family that merely copied the wire-EDM thresholds would be worse than none.

MEANWHILE None. Sinker EDM is not offered. The two processes it competes with, wire EDM and milling, are both in the catalogue, and a blind sharp-cornered pocket now surfaces as a wire-EDM finding ("blind features cannot be wire-cut") that names sinker EDM in its fix.

08 Tube bending — bend radius against tube OD, and wall thinning round the bend

The part recognised AS a tube: a circular section swept along a centreline, so the outside diameter, the wall and the centreline radius can be read separately. The engine finds cylinders and it finds bends between planar flanges; it has no swept-section recogniser, and a bent tube is neither a body of revolution nor folded sheet.

MEANWHILE None. Tube Bending is the ONE shaping process in this tool priced and carbon-scored but not judged, and the picker prints that reason when it is selected rather than showing an empty report.

09 Corner ANGLE for the fine-blanking corner-radius rule

The included angle at each internal corner. The published guidance is angle-dependent — an obtuse corner needs 5-10% of thickness, a right angle 10-15%, an acute one 25-30% — and the catalogue applies the single 0.5t "minimum inside radius" figure to every corner because the angle is not measured.

MEANWHILE `fb-corner-radius` uses the flat 0.5t figure, which is stricter than the obtuse case and more permissive than the acute one. A sharp acute corner can therefore pass a rule it should fail.

10 True LIMITING DRAW RATIO (blank diameter / punch diameter)

The blank and the punch. Neither is in a finished solid model: the blank is the flat sheet before forming and the punch is tooling. Computing the real LDR would need the part unfolded to its developed blank, which is a forming simulation, not a measurement.

MEANWHILE `dd-draw-depth` reads bounding-box depth over the narrower box width, which is the depth-to-diameter figure design guides quote alongside the LDR. It cannot see a stepped or tapered cup, where the governing diameter is not the box width, and it ABSTAINS when the draw direction is not a box axis.

11 Minimum as-cast cored hole diameter for SAND casting

A published sand-foundry figure for the smallest hole worth coring. Every other casting family in this catalogue now carries one — HPDC 2.5 mm, zinc 1.5 mm, permanent mould 6.0 mm, investment 1.5 mm — and sand does not, because the research turned up a sand CORE cross-section floor (about 1.6 mm for stability against the melt) and an unsupported core L/D band (10-15), but no sand cored-HOLE minimum from any source. Interpolating one from the permanent-mould 6 mm would have looked identical on the page to the four that are sourced.

MEANWHILE `sand-core-ld` still catches a sand core that is too SLENDER. Nothing catches one that is simply too fine, so a small hole on a sand casting is currently unjudged rather than passed — it appears under NOT EVALUATED with this reason.

12 Vacuum-assisted die casting as a distinct GEOMETRIC family

Nothing — and that is the point. Evacuating the cavity changes the gas in it, not the shape the die can make: wall, draft, core slenderness and undercut limits are the die-casting limits either way. What vacuum buys is porosity low enough to heat-treat and weld, which is a metallurgical property this tool does not measure. The process is offered in the picker and priced separately, and routed to the HPDC rule family on purpose.

MEANWHILE The `hpdc` family judges it, correctly. A near-copy family would have invented six thresholds to restate the same limits and inflated the catalogue count by six.

13 Tool HOLDER and machine-envelope collision (machining)

The holder geometry, the spindle nose and the machine work envelope. The shank-clearance sweep behind `mach-tool-access` now measures whether a cutter of a given diameter can reach each face along each approach direction, which is the larger half of this — but a face it calls reachable can still be unreachable once the holder is on the tool.

MEANWHILE `mach-tool-access` measures shank clearance and reports it as a LOWER bound on the access problem.

14 Tolerance STACK-UP along a dimensional chain

The datum reference frames and the chain of dimensions between them, so accumulated variation can be propagated feature to feature. Reading individual callouts is not the same problem: a part can pass every tolerance on the drawing and still fail at assembly because they add up.

MEANWHILE Individual dimensions, geometric tolerances and datums ARE read from semantic AP242 PMI, and the tightest callout is checked against each process's capability. What is missing is the accumulation, not the reading.

15 **Datum reference frames (AP242 PMI)**

The link from each geometric tolerance to the datums it is measured from. The reader extracts datum labels, but on the round-trip fixture written by this repo none survived the write, so the datum path is IMPLEMENTED AND UNVERIFIED — it has never been proved against a file with datums in it.

MEANWHILE Datum COUNT is reported and is zero on every file tested so far, which is honest but untested. Treat a non-zero datum count from a vendor CAD export as unproven until a fixture confirms it.

16 **Sink and warp prediction (injection moulding)**

A mould-flow simulation: fill pattern, cooling layout and differential shrinkage.

MEANWHILE Wall uniformity and rib proportions cover the two geometric causes the engine CAN see.

17 **Blank nesting utilisation (sheet metal)**

A flat pattern and a strip layout. Unfolding a formed part is a solver in its own right and the recogniser does not do it.

NO PROXY - THE ENGINE REPORTS NOTHING IN THIS AREA

18 **Press tonnage and station count (sheet metal)**

Cut perimeter from the flat pattern, plus a progressive-die strip layout. The same unfolding gap as nesting.

MEANWHILE stampingFeatureCost estimates forming cost from the recognised bend count, which is a cost output rather than a design rule.

APPENDIX B

Threshold audit register

The catalogue grades its citations, but a grade is a claim about a source rather than a record that anybody opened it. This is that record. 38 of 247 thresholds have been through it so far; the rest are marked NOT YET AUDITED on their own entries, and that is the honest state of them. Run "node scripts/threshold-audit.mjs" for the live work list.

CONTESTED

Wall thickness outside the die-casting range

`hpdc-wall-thickness-range` · High-pressure die casting (Al / Mg)

STILL CONTESTED, and the primary document confirms why rather than settling it. Product Design for Die Casting says plainly in §3.1: 'There are no hard and fast rules governing maximum and minimum limits for wall thicknesses.' It gives no table. What it DOES give is the physical argument, from §4.2: the dense chilled skin is 0.38-0.50 mm and is NOT a function of wall thickness, so a thick wall carries the same load-bearing skin as a thin one plus a porous core - which makes a heavy section useless rather than merely inefficient. Table 4.8 separately rates each aluminium alloy for die-filling capacity (1 best to 5 worst), which is the one axis on which the minimum wall genuinely varies.

RECOMMENDATION Keep the band at industry-consensus and do NOT promote it. The rationale should carry the skin argument, and the minimum should scale with the alloy's die-filling rating rather than being flat across every aluminium grade.

CONTESTED

Punched hole smaller than the material thickness

`sm-hole-diameter` · Sheet metal / stamping

GENUINE CONFLICT on the high-strength grades. Sec. 9.3.3(a) sizes the punch so it does not crush: combined with the punch force of Eq. 4.3 this gives d_min/T = 2.8*UTS/sigma_pd, with sigma_pd = 980-1560 MPa for hardened tool steel. At the conservative end that is 2.80 d/t for DP980 against the 1.50 d/t the catalogue holds, 1.77 for 304 against 1.20, and 1.66 for 316L against 1.20 — the book is stricter on every high-strength grade. It is LOOSER on 22MnB5 (1.71 vs our 2.50), but our figure there is for the hardened condition, which the book does not cover. The book's number is first-principles; ours is unattributed prose.

RECOMMENDATION Do not switch silently. The engine now computes the book's limit alongside ours and publishes both (`punchedHoleMarginRatio` and `_punchLimit`); a plant that pierces DP980 at 1.5 d/t successfully is evidence the 980 MPa punch-stress floor is too conservative for their tooling, and that is exactly the measurement needed to settle it. Resolve with one real part before changing either number.

PRIMARY SOURCE READ

Wall area below the draft NADCA computes for this alloy and depth

`hpdc-draft-minimum` · High-pressure die casting (Al / Mg)

RESOLVED FROM THE PRIMARY DOCUMENT. The user supplied the book itself, so the blocker that made this CONTESTED is gone. NADCA does not give a draft constant at all - it gives a formula, A = 57.2738 / (C*sqrt(L)) with L the feature depth in inches, and a table of C by alloy group and surface: zinc/ZA 50 inside / 100 outside / 34 hole total, magnesium 35/70/24, aluminium 30/60/20, copper 25/50/17. The formula was verified against the book's own worked example for an aluminium inside surface (0.1 in -> 6.0 deg, 1.0 in -> 1.9 deg, 5.0 in -> 0.85 deg) before being implemented. The old flat 1 degree was correct at about 23 mm of depth and wrong everywhere else: on a 6 mm deep pocket the book asks 1.96 deg on an outside surface and 3.93 deg on an inside one. The rule now computes the requirement from the alloy and the part's own measured depth along the draw.

RECOMMENDATION The INSIDE half of the split is still not implemented. The rule judges every wall at the OUTSIDE constant, which is the lower of the two, so it under-reports on ejector-half surfaces by a factor of two. That is stated in the rule's own source text rather than left for a reader to discover. Closing it needs a solid-side test that survives through-features - from a bore wall the outward normal escapes through the bore opening, so the obvious ray test classifies a through-hole as an outside surface.

PRIMARY SOURCE READ

Cored hole below the draft NADCA computes for its depth

`hpdc-cored-hole-draft` · High-pressure die casting (Al / Mg)

RESOLVED FROM THE PRIMARY DOCUMENT. The book gives a cored-hole constant per alloy group - aluminium 20, magnesium 24, zinc and ZA 34, copper 17 - and the column is the TOTAL included angle, so the per-side requirement is half of what the formula returns. The old flat 2 degrees per side matches the book at about 13 mm of bore depth and nowhere else: a 3 mm bore in aluminium needs 4.17 deg per side and a 50 mm bore needs 1.02. The rule now derives each bore's depth from its own measured wall area and radius and judges the worst bore on the part.

RECOMMENDATION None outstanding. The value, the alloy split and the total-versus-per-side distinction all come from the document as printed.

PRIMARY SOURCE READ

Inside bend radius tighter than the grade will take

sm-bend-radius · Sheet metal / stamping

CORROBORATED, not superseded. The catalogue's twelve per-alloy values were checked row by row against Table 5.2 (R_min = c*T, c by material group and temper). Every steel and stainless value we hold sits INSIDE the book's soft-to-hard band, and 7075 at 4.0 r/t sits inside the 1.5-6.0 band for high-strength heat-treatable aluminium. The book does NOT supply our values — it has no row for 6000-series aluminium and predates dual-phase steel entirely — so the catalogue's figures stand and the book is the check on them, not the source.

RECOMMENDATION Keep the per-alloy table. The book adds a dimension we do not model at all — TEMPER — with c ranging 0.5 to 3.0 for the same low-carbon steel. Capturing temper as an input would let the rule tighten or relax correctly instead of holding one figure per alloy.

PRIMARY SOURCE READ

Bend radius too LARGE to take a permanent set

sm-bend-radius-max · Sheet metal / stamping

R_i(max) <= T*E/(2*YS). A bend gentler than this never yields and the part springs back flat. New capability — nothing in the catalogue tested for it.

RECOMMENDATION None outstanding. The source was read first-hand and the equation is implemented as printed.

PRIMARY SOURCE READ

Springback beyond what overbend alone can compensate

sm-springback · Sheet metal / stamping

Springback factor from the material's own YS/E, with the 2-8% practical overbend allowance stated in the text. New capability — springback was entirely absent.

RECOMMENDATION None outstanding. The source was read first-hand and the equation is implemented as printed.

PRIMARY SOURCE READ

Strip utilisation below the industry target

sm-blank-utilisation · Sheet metal / stamping

Webs m = T + 0.015*D and n from Table 4.3, against the text's own 70-80% utilisation target. The utilisation figure is a LOWER bound: it assumes a rectangular envelope in a single-pass layout (Fig. 4.7a) and real nesting does better.

RECOMMENDATION None outstanding. The source was read first-hand and the equation is implemented as printed.

PRIMARY SOURCE READ

Cup too deep for the drawing-ratio table

dd-draw-stages · Deep drawing (multi-stage)

Successive drawing ratios m1..m5 by relative thickness 100*T/D. The blank diameter is estimated by area equivalence rather than the book's Guldinus method (Sec. 6.4.2), which needs a meridian profile the recogniser does not extract — so the STAGE COUNT is sound and the blank it starts from is an estimate.

RECOMMENDATION None outstanding. The source was read first-hand and the equation is implemented as printed.

PRIMARY SOURCE READ

Inside corner radius smaller than the wall it joins

hpdc-internal-radius · High-pressure die casting (Al / Mg)

RESOLVED FROM THE PRIMARY DOCUMENT, and the old value was wrong in kind rather than in degree. The rule held a flat 1.6 mm. NADCA gives the fillet as a RATIO to the wall it joins: R1 >= T1 at a junction of uniform walls with R2 = R1 + T1, and R1 >= 2/3(T1 + T2) between non-uniform walls; on rib tops and tee junctions R1 = 0.5*T1 is optimal and below 0.25*T1 is 'not recommended'. A flat 1.6 mm passes a 1.6 mm radius on a 6 mm wall where the book wants 6 mm, so the rule was silent on exactly the chunky castings it should have been loudest about.

RECOMMENDATION The outside corner radius R2 = R1 + T is specified by the book and is not yet checked, because the recogniser reports inside corner radii only. Adding external corner recognition would close it.

PRIMARY SOURCE READ

Tolerance tighter than NADCA Standard capability at that dimension

hpdc-tolerance-capability · High-pressure die casting (Al / Mg)

RESOLVED FROM THE PRIMARY DOCUMENT, supplied by the user as the 2021 11th edition. The blocker this register carried - 'NADCA Product Specification Standards #402 not held' - is closed. The standard does not give one screening value: linear capability is a function of the dimension (Al/Mg/Zn Standard +/-0.25 mm for the first 25.4 mm plus +/-0.025 mm per additional inch; Precision +/-0.05 mm base; copper +/-0.36/+/-0.076 and +/-0.18/+/-0.05). The rule now computes capability per PMI dimension at its own size, or judges a declared band against the first-inch base with that basis stated. Implementation verified against the standard's own worked examples: 127

mm aluminium -> +/-0.35 mm Standard, +/-0.15 mm Precision; the cross-parting-line composition reproduces +0.65/-0.35 mm.

RECOMMENDATION Cross-parting-line and moving-die adders (S-4A-2/3) are computed for the report but not yet attributed to individual dimensions - that needs to know which PMI dimension crosses the parting line, which the draw direction plus the dimension's axis could supply later.

PRIMARY SOURCE READ

Tolerance tighter than NADCA Standard capability at that dimension

`hpdc-zinc-tolerance-capability` · `High-pressure die casting (Zinc)`

RESOLVED FROM THE PRIMARY DOCUMENT. Zinc shares the aluminium row at Standard grade (+/-0.25 mm base). The P-4A-1 notes record that artificial ageing and Section 4B miniature machines reach tighter than the table - which is why the old flat 0.15 mm screen was neither right nor wrong: the standard's own answer is dimension-dependent.

RECOMMENDATION Cross-parting-line and moving-die adders (S-4A-2/3) are computed for the report but not yet attributed to individual dimensions - that needs to know which PMI dimension crosses the parting line, which the draw direction plus the dimension's axis could supply later.

PRIMARY SOURCE READ

Flatness tighter than the as-cast table holds

`hpdc-flatness-capability` · `High-pressure die casting (Al / Mg)`

NEW RULE from the primary document: as-cast flatness, all alloys, 0.20 mm to 76.2 mm largest dimension + 0.08 mm per additional inch (Standard); 0.13 + 0.05 (Precision). Verified against both printed worked examples (254 mm diagonal -> 0.76 / 0.48 mm).

RECOMMENDATION Declared-input only: a STEP file does not say which face must be flat. PMI flatness callouts could feed it once the extractor reports them per-face.

PRIMARY SOURCE READ

Flatness tighter than the as-cast table holds

`hpdc-zinc-flatness-capability` · `High-pressure die casting (Zinc)`

NEW RULE from the primary document: as-cast flatness, all alloys, 0.20 mm to 76.2 mm largest dimension + 0.08 mm per additional inch (Standard); 0.13 + 0.05 (Precision). Verified against both printed worked examples (254 mm diagonal -> 0.76 / 0.48 mm).

RECOMMENDATION Declared-input only: a STEP file does not say which face must be flat. PMI flatness callouts could feed it once the extractor reports them per-face.

PRIMARY SOURCE READ

Tolerance tighter than sand casting holds

`sand-tolerance-capability` · `Sand casting`

RESOLVED FROM THE PRIMARY DOCUMENT for steel. The user supplied the supplement: the ISO 8062-1994 CT table is adopted as SFSA 2000 (statistical basis: 140,000+ features on production steel castings), with grade selection CT11-13 typical for a short production series (judged at CT13) and CT10-12 for a long one (judged at CT12). The mm table was verified against the printed inch table before wiring. The old flat 1.2 mm screen was TIGHTER than anything the tables promise for steel - CT13 never tabulates below 6 mm - meaning the screen passed bands no steel foundry could hold as-cast. Non-ferrous sand alloys keep the screening value: the CT value table is metal-independent in ISO 8062, but the grade selection for non-ferrous is not held.

PRIMARY SOURCE READ

Tolerance tighter than investment casting holds

`inv-tolerance-capability` · `Investment casting`

RESOLVED for steel: investment castings CT5-7 (long production series only - investment tooling is always iterated), judged at CT7. The old 0.3 mm screen sat close to the CT5-7 values at small dimensions, so verdicts move little on small parts and become dimension-aware on large ones. Non-steel investment alloys keep the screen.

PRIMARY SOURCE READ

Local section below the process minimum

`sand-min-section` · `Sand casting`

RESOLVED FROM THE PRIMARY DOCUMENT for the steel band. The user supplied the supplement itself: p.2 prints "a minimum thickness of 0.25 in. (6 mm) is suggested for design use when conventional steel casting techniques are employed", and Fig. 1 raises the minimum with the length the molten steel must run - digitized at +/-1.5 mm, and used sharpen-only at the part's own longest dimension. The ferrous family band moved from a researched 5.0 mm to the printed 6.0 mm. The generic 3.5 mm band and the castiron/copper bands keep their prior grading - this document speaks for steel only.

PRIMARY SOURCE READ

Rib heavier than the wall it stands on

sand-rib-thickness-max · Sand casting

RESOLVED for the steel band: the thermally-neutral rib proportions of Fig. 8 stop at T1 = 3/4 T2, so the ferrous band is 0.75 where the generic band stays 1.0. The full coupling (height limit moving with thickness ratio: 4 walls at 1/4, 1.5 at 1/2, 0.5 at 3/4) is enforced by the dedicated sand-rib-thermal-neutrality rule.

PRIMARY SOURCE READ

Steel rib taller than its thermally neutral height

sand-rib-thermal-neutrality · Sand casting

NEW RULE from the primary document: rib height limit COUPLED to rib thickness ratio, interpolated between the three printed points, worst rib named. Steel castings only; abstains on every other alloy.

PRIMARY SOURCE READ

Junction fillet below the steel-casting band

sand-junction-fillet · Sand casting

NEW RULE from the primary document: junction fillet r = T1 clamped to the printed 13-25 mm band; the cap is deliberate ((D/d)^2 mass growth at the joint). Steel castings only.

PRIMARY SOURCE READ

Boss more than twice the wall it stands on

sand-boss-dia · Sand casting

NEW RULE from the primary document: the boss-neutrality bands stop at D = 2.0 T, beyond which no neutral height is printed. Worst boss sets the ratio. Steel castings only.

PRIMARY SOURCE READ

Flatness callout tighter than sand casting holds

sand-flatness-capability · Sand casting

NEW RULE: flatness by ISO 8062-2 CTG grade at the bounding diagonal; steel sand machine-molded CTG5-7, judged at CTG7 (hand molding extends to CTG8, stated in the basis). Abstains without a declared callout, and on non-steel alloys.

PRIMARY SOURCE READ

Flatness callout tighter than investment casting holds

inv-flatness-capability · Investment casting

NEW RULE: steel investment CTG4-6, judged at CTG6 at the bounding diagonal. Abstains without a declared callout, and on non-steel alloys.

PRIMARY SOURCE READ

Machining stock below the required machining allowance

sand-machining-stock · Sand casting

NEW RULE: declared machining stock vs the required machining allowance at the casting's largest dimension, judged at grade F - the machine-molded band's minimum, the least the standard ever requires. The printed 6 mm cope-surface floor is stated, not silently applied (orientation is the foundry's choice). Steel only; abstains undeclared.

PRIMARY SOURCE READ

Wall area below the minimum draft angle

im-draft-minimum · Injection moulding

RESOLVED for the resins Table 3.01 names: the draft ANGLE is now computed per resin and draw depth (Zytel 1/8-1/2 deg, Delrin 1/4-1/2, reinforced nylons and Rynite 1/2-1, judged at the top of each printed range at the part's own extent along the draw - the table asks more of deeper draws, so the part extent is conservative). This REPLACES the generic 1 deg for named resins - nylon genuinely needs less, and the primary source saying so is the point. ABS/PC/PP and other unnamed resins keep the generic angle and grade. Texture (+1 deg per 0.001 in) is recorded; texture depth is not yet a declared input. SECOND PRIMARY SOURCE: Covestro (Bayer) "Engineering Polymers: Part and Mold Design - Thermoplastics, A Design Guide", p.32 - PC-based materials at least 0.5 deg per side (1 deg preferred), Apec high-heat PC and TPU prefer 3-5 deg (judged at 3). Flat per resin, no printed depth dependence; the same texture rule of thumb as DuPont.

PRIMARY SOURCE READ

Rib too thick at its base for the wall it stands on

`im-rib-thickness-max` · Injection moulding

CONFIRMED, not retuned: the printed band (0.4 W appearance / 0.6 W structure) matches the tool's existing 0.4/0.6 thresholds exactly. Grade upgraded to standard-named; the stricter resin-specific bands (0.5 for high-shrink semi-crystallines) are practising-moulder experience, stand unchanged, and carry their own industry-consensus grade. SECOND PRIMARY SOURCE (read 2026-08-11): Covestro (Bayer) "Engineering Polymers: Part and Mold Design - Thermoplastics, A Design Guide", Table 2-1 p.25 - the digit font in the supplied copy does not extract, so the table was read from the page image. It TIGHTENS the PC ceiling from the 0.6 consensus to the maker's printed 0.5 (minimal sink; 0.4 high gloss; 0.66 where slight sink is acceptable) and adds Bayblend PC/ABS and Texin/Desmopan TPU at 0.5, all standard-named.

PRIMARY SOURCE READ

Rib too thin to fill reliably

`im-rib-thickness-min` · Injection moulding

CONFIRMED at the printed 0.4 W appearance floor; grade upgraded. Resin entries (PP 0.3, glass-filled nylon 0.5) stand on their own grade.

PRIMARY SOURCE READ

Inside corner radius below the DuPont stress-concentration knee

`im-internal-radius` · Injection moulding

NEW RULE: fillet radius at least half the wall (the printed stress-concentration knee at R/T = 0.5) with the printed 0.508 mm floor. DuPont-named resins only; abstains on others. SECOND PRIMARY SOURCE: Covestro (Bayer) "Engineering Polymers: Part and Mold Design - Thermoplastics, A Design Guide", p.31 + Fig 2-22 - r/t of approximately 0.15 for PC-family resins, a DIFFERENT claim from DuPont's 0.5 knee (compromise incl. appearance vs pure stress flattening); each resin is judged by the book that names it, gates disjoint by construction and pinned by test.

PRIMARY SOURCE READ

Boss outside diameter outside the 2–2.5× hole band

`im-boss-od` · Injection moulding

NEW RULE: boss OD 2-2.5x its hole, two-sided (weak below, sink/void mass above), judged on the worst coaxial boss/hole pair; abstains when none is recognised.

PRIMARY SOURCE READ

Blind hole deeper than twice its diameter

`im-blind-core-depth` · Injection moulding

NEW RULE: blind holes limited to twice their diameter (one-ended core pin deflection); through holes are exempt because the pin is supported at both ends. Resin-independent.

PRIMARY SOURCE READ

Boss too tall for the core pin that forms it

`im-boss-height` · Injection moulding

PARTIALLY RESOLVED with a genuine two-claim conflict handled sharpen-only: Covestro prints 5x OD as the point where FILLING becomes the problem; the existing PC screen of 3x rests on notch-sensitivity (crazing at the boss base under load) - a different failure mode and STRICTER, so it stands with both claims named in the source. PC/ABS and TPU (no notch screen) are judged at the printed 5x, standard-named. Generic 3x stays for unnamed resins.

CORROBORATED BY SEARCH

Cored hole beyond the core-pin slenderness limit

`hpdc-core-ld` · High-pressure die casting (Al / Mg)

Widely published as a practical limit; the exact figure varies with alloy, pin cooling and whether the pin is supported at both ends. A single number across all cored holes is a simplification the report should keep stating.

RECOMMENDATION Where a customer standard exists it should override this, which the threshold resolver already supports.

BLOCKED BY: NO PRIMARY SOURCE REACHABLE FROM THIS ENVIRONMENT

NOT YET AUDITED

Cored holes below the core draft

`gdc-cored-hole-draft` · Gravity die casting

Derived by the same reasoning as the aluminium cored-hole rule — a cored hole needs more draft than a wall because the casting shrinks onto the core — at 3.0 deg for this family. NOT independently corroborated: it is consistent with the family's own wall rule and with the aluminium figure, which is an argument, not evidence.

RECOMMENDATION Corroborate against a permanent-mould foundry design guide before relying on it.

NOT YET AUDITED

Cored holes below the core draft

`lpdc-cored-hole-draft` · Low-pressure die casting

Derived by the same reasoning as the aluminium cored-hole rule — a cored hole needs more draft than a wall because the casting shrinks onto the core — at 2.0 deg for this family. NOT independently corroborated: it is consistent with the family's own wall rule and with the aluminium figure, which is an argument, not evidence.

RECOMMENDATION Corroborate against a permanent-mould foundry design guide before relying on it.

NOT YET AUDITED

Cored hole below the draft NADCA computes for zinc at its depth

`hpdc-zinc-cored-hole-draft` · High-pressure die casting (Zinc)

Derived by the same reasoning as the aluminium cored-hole rule — a cored hole needs more draft than a wall because the casting shrinks onto the core — at 1.0 deg for this family. NOT independently corroborated: it is consistent with the family's own wall rule and with the aluminium figure, which is an argument, not evidence.

RECOMMENDATION Corroborate against a permanent-mould foundry design guide before relying on it.

NOT YET AUDITED

Tolerance tighter than low-pressure die casting holds

`lpdc-tolerance-capability` · Low-pressure die casting

NOT covered by NADCA #402: its tables carry the four conventional die-casting alloy columns only. The primary source for permanent-mould, sand and investment tolerance grades is ISO 8062-3 (dimensional and geometrical tolerances for moulded parts - CT grades), which is not held.

RECOMMENDATION Obtain ISO 8062-3. Until then these stay unaudited industry consensus.

BLOCKED BY: ISO 8062-3 NOT HELD; NADCA #402 (NOW HELD) DOES NOT TABULATE THESE PROCESSES

NOT YET AUDITED

Tolerance tighter than gravity die casting holds

`gdc-tolerance-capability` · Gravity die casting

NOT covered by NADCA #402: its tables carry the four conventional die-casting alloy columns only. The primary source for permanent-mould, sand and investment tolerance grades is ISO 8062-3 (dimensional and geometrical tolerances for moulded parts - CT grades), which is not held.

RECOMMENDATION Obtain ISO 8062-3. Until then these stay unaudited industry consensus.

BLOCKED BY: ISO 8062-3 NOT HELD; NADCA #402 (NOW HELD) DOES NOT TABULATE THESE PROCESSES

NOT YET AUDITED

Tolerance tighter than injection moulding holds

`im-tolerance-capability` · Injection moulding

DuPont Module I (now held, read 2026-08-11) prints NO tolerance tables - it is a design-principles book. The moulding tolerance primary source remains DIN 16742 (successor of DIN 16901), not held; the SPI/SPE commercial-vs-fine charts would also serve.